Wildfires and Birth Outcomes: Evidence from Lightning Strikes*

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Abstract

This study estimates the causal effect of wildfires on health at birth by using lightning strikes as an instrumental variable. In the birth month, a one standard deviation increase in natural-caused wildfire size from its sample mean increases the incidence of low birth weight by 1.1 per 1,000 births. Alongside the existing evidence on wildfire smoke, the results provide suggestive evidence of a local-disruption channel, including stress-related responses. Finally, the estimated adverse birth-weight effect is smaller in counties where more land had previously been treated to reduce wildfire fuels through prescribed burning or mechanical vegetation removal.

JEL Classification: Q53, Q54, I12, J13
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1 Introduction

Climate change is increasing the frequency and severity of natural disasters worldwide. Growing evidence documents that extreme weather events impose substantial and lasting costs on public health and economic activity (e.g., Dell et al., 2014; Carleton and Hsiang, 2016). Among climate-driven hazards, wildfires are a particularly important threat because they combine local disaster risk with large-scale air pollution. Climate change increases fuel aridity and creates fire weather conditions that make large wildfires more likely (Abatzoglou and Williams, 2016). Between 2003 and 2020, 1,431,977 wildfires occurred in the contiguous US, burning 423,580.9 km^2 (Short, 2022). Wildfires account for approximately 25% of total fine particulate matter ($PM_{2.5}$) emissions in the US (Burke et al., 2021), and wildfire-prone areas have experienced deteriorating $PM_{2.5}$ trends even as air quality has improved elsewhere (McClure and Jaffe, 2018).

These trends raise concerns about the growing health burden of wildfires. Since wildfires release large amounts of smoke into the atmosphere, a growing literature documents adverse effects of wildfire smoke on health (e.g., Heft-Neal et al., 2023; Molitor et al., 2023; Miller et al., 2024). Wildfires may also threaten health through psychological stress associated with evacuation, threats to homes and livelihoods, and disruption of daily life.[1] Although the health consequences of wildfires may extend beyond smoke exposure, little is known about the causal effect of local wildfire events on health.

The objective of this study is to estimate the causal effect of wildfires on health, focusing on local wildfire events rather than transported smoke exposure. The empirical challenge is that wildfire occurrence and size are endogenous because many wildfires are caused by human activity. Human-caused wildfires can be correlated with local economic activity, agricultural income, crime, land management, and residential sorting. For example, agricultural fires may be positively associated with birth outcomes because income gains associated with agricultural production can confound estimates of fire-related health damages (Rangel and Vogl, 2019).

To address this endogeneity, this study exploits plausibly exogenous variation in natural-caused wildfire size generated by lightning strikes. Lightning is one of the main sources of natural-caused wildfire ignition, and conditional on county fixed effects, year-month fixed effects, and detailed fire weather controls, the timing and location of lightning strikes provide quasi-random variation in wildfire size. This study uses lightning strikes as an instrumental variable (IV) for natural-caused wildfire size in a county-month panel covering 2003–2020. The analysis combines incident-level wildfire records from the US Forest Service (USFS), lightning data, wildfire-smoke measures, fire-damage records, fuel-treatment data, and county-month natality records from the Centers for Disease Control

[1] Studies of hurricanes and earthquakes document adverse effects of prenatal disaster exposure on infant health and suggest maternal stress as a potential mechanism (e.g., Currie and Rossin-Slater, 2013; Kim et al., 2017; de Oliveira et al., 2023).

and Prevention (CDC). The baseline sample covers counties with more than 100,000 residents, representing approximately 60% of the US population. The main outcomes of interest are birth outcomes, which are sensitive to environmental insults and predict later-life human capital (e.g., Currie et al., 2009; Almond and Currie, 2011; Currie et al., 2014).

Using wildfire records on ignition causes, the ordinary least squares (OLS) analysis shows positive associations between human-caused wildfire size in the birth month and birth outcomes. These associations are consistent with the endogeneity concerns discussed above. Human-caused wildfires may reflect local socioeconomic conditions that also affect infant health. In contrast, the IV estimates using the number of lightning strikes as an instrument indicate that natural-caused wildfires in the birth month worsen birth outcomes. In the birth month, a one standard deviation increase (approximately 12.6 km^2) in natural-caused wildfire size from its sample mean reduces birth weight by 4.1 grams and increases the incidence of low birth weight by 1.1 per 1,000 births. The IV results using the obstetric estimate of gestational age also indicate shorter gestation and a higher incidence of preterm birth.

Much of the existing wildfire-health literature measures exposure using Hazard Mapping System (HMS) smoke plumes or wind-driven smoke transport (Fadadu et al., 2021; Borgschulte et al., 2024; Heft-Neal et al., 2022; Molitor et al., 2023; Cabral and Dillender, 2024; Miller et al., 2024). The IV estimates show that larger natural-caused wildfires increase both HMS smoke coverage and wildfire-specific $PM_{2.5}$ concentrations. Local wildfire events may also affect birth outcomes through evacuation risk, property damage, and psychological stress. To assess the role of smoke exposure in the estimated effects, this study adds controls for HMS smoke coverage and wildfire-specific $PM_{2.5}$ (Childs et al., 2022). These controls do not materially change the estimated effects of wildfire size on birth outcomes.

Furthermore, this study examines Twitter sentiment and maternal tobacco use as additional outcomes of local wildfire exposure. The estimates indicate that natural-caused wildfires lower Twitter sentiment and increase maternal tobacco use. Another outcome is the number of buildings involved in a wildfire, which is a proxy for local disruption. Natural-caused wildfires cause greater reported building involvement and per-resident property losses. Together, these patterns are consistent with effects of local disruption beyond measured smoke exposure, including stress-related responses.

This study also examines heterogeneity in the health effects of wildfires by maternal age, finding larger adverse effects on birth weight and low birth weight in county-months with younger average maternal ages. This pattern is consistent with evidence from Hurricane Catarina in Brazil, where adverse effects on birth outcomes are concentrated among mothers aged 15–24 (de Oliveira et al., 2023).[2] Taken together, the auxiliary-outcome and

[2]The estimates also indicate larger adverse effects on birth weight and low birth weight in county-

heterogeneity results suggest a role for maternal stress in the adverse effects of wildfires on birth outcomes, but do not rule out effects operating through air pollution.

A final set of analyses examines fuel treatment as a potential mitigating factor. Fuel treatments, including mechanical removal and prescribed fire, are a primary adaptation tool for reducing wildfire risk (Bayham et al., 2022; Strabo and Reimer, 2024; Strabo et al., 2026). This study constructs a measure of cumulative fuel-treatment area within each county, then examines whether the effects of lightning-induced wildfire size vary with this preexisting fuel-treatment area. The estimates indicate a smaller adverse birth-weight effect in counties with a larger fuel-treatment area. Thus, fuel treatment may mitigate the health effects of wildfires, which is consistent with evidence that fuel treatment reduces wildfire size and severity (Strabo and Reimer, 2024; Strabo et al., 2026).

The central contribution of this study is to provide a lightning-based research design for estimating the health effects of natural-caused wildfires. Existing quasi-experimental studies largely identify wildfire-smoke effects using satellite smoke measures or wind-driven smoke transport (e.g., Jayachandran, 2009; Borgschulte et al., 2024; Molitor et al., 2023; Cabral and Dillender, 2024; Miller et al., 2024). The most closely related study, by Heft-Neal et al. (2022), focuses on preterm birth and documents an increased risk associated with prenatal wildfire smoke exposure. This study examines local wildfire events and additionally shows that natural-caused wildfires reduce birth weight. Related evidence on mental health comes from Currie and Saberian (2025), who examine smoke exposure alongside nearby fires, evacuation, and wildfire-related anxiety. This study complements that work by estimating the effects of local wildfire events on birth outcomes while addressing confounding from human ignitions, land use, and local economic activity.

Beyond this central contribution, the analysis offers two complementary extensions. First, the smoke-control, auxiliary-outcome, and heterogeneity estimates provide suggestive evidence of a local-disruption channel that includes stress-related responses. This evidence complements research on the effects of natural disasters on birth outcomes through maternal stress (e.g., Currie and Rossin-Slater, 2013; Kim et al., 2017; de Oliveira et al., 2023) and suggests that wildfire-health effects may extend beyond smoke exposure. Second, by examining local wildfire events and the heterogeneity of health effects by preexisting fuel-treatment area, this study suggests that fuel treatment can mitigate the health effects of wildfires. These findings are relevant to the growing literature on wildfire adaptation and mitigation (Strabo and Reimer, 2024; Strabo et al., 2026).

The remainder of this paper is organized as follows. Section 2 provides a brief background on wildfires in the US and the potential impact of wildfires on air pollution and health. Section 3 describes the data used in this study. Section 4 presents the empirical

months with higher Black maternal shares. Evidence on racial heterogeneity in birth outcomes is mixed in studies of both air pollution (e.g., Currie and Walker, 2011; Heft-Neal et al., 2022) and natural disasters (e.g., Currie and Rossin-Slater, 2013; Mark et al., 2025). These interactions capture differences across county-month maternal demographic profiles rather than individual-level subgroup effects.

strategy in detail. Section 5 reports the estimation results. Section 6 evaluates the policy implications of fuel treatment as a wildfire-mitigation tool. Section 7 concludes.

2 Background

2.1 Wildfires in the US

According to the US Department of Agriculture (USDA), the term wildfire refers to "unplanned fires that occur in wildlands such as forests, rangelands or grasslands."[3] Wildfires are widespread across the US. In the contiguous US, the number of wildfires from 2003 to 2020 totaled 1,431,977, and the total area burned was 423,580.9 km^2 (Short, 2022). Relative to the approximately 3.1 million km^2 of forest area in the US,[4] this burned area represents a substantial environmental and economic exposure.

Table 1: Summary Statistics for Wildfire Size by Ignition Cause, 2003–2020

	Obs.	Mean	SD	Max	Sum
Arson/incendiarism	171,081	0.1	3.1	649.0	22,128.2
Debris and open burning	342,628	0.1	1.6	658.4	20,283.3
Equipment and vehicle use	122,175	0.2	8.3	1,660.0	28,919.8
Firearms and explosives use	2,346	1.0	10.4	331.1	2,317.7
Fireworks	13,849	0.1	2.8	196.2	1,406.8
Missing data/not specified/undetermined	416,597	0.3	10.7	2,681.9	106,428.9
Misuse of fire by a minor	33,335	0.0	1.3	155.2	1,113.1
Natural (including lightning)	188,185	1.1	20.2	2,385.1	210,433.7
Other causes	7,259	0.2	6.5	365.4	1,507.5
Power generation/transmission/distribution	28,150	0.4	9.3	801.2	12,012.2
Railroad operations and maintenance	12,230	0.1	1.0	57.5	1,010.4
Recreation and ceremony	60,558	0.2	11.9	2,177.4	14,330.8
Smoking	33,584	0.1	2.0	236.7	1,688.5
TOTAL	1,431,977	0.3	10.2	2,681.9	423,580.9

Notes: This table reports unweighted summary statistics for wildfire size (km^2) by cause. The observational unit is the wildfire event. The columns report the number of observations (Obs.), mean (Mean), standard deviation (SD), maximum (Max), and total (Sum).

The USFS database distinguishes natural causes, human causes, and undetermined or unclassified causes. Human-caused wildfires include open burning and arson, while natural-caused wildfires are mainly caused by lightning strikes that ignite a tree or other fuel source. Table 1 reports the summary statistics for wildfire size by cause. By number of incidents, a large share of wildfires have an undetermined or unclassified cause (29.1%). Among identified causes, "debris and open burning" and "arson/incendiarism" are the most frequent categories, accounting for approximately 23.9% and 11.9% of all incidents,

[3]USDA: https://www.climatehubs.usda.gov/taxonomy/term/398

[4]DataBank, World Bank: http://databank.worldbank.org/data/reports.aspx?source=2&type=metadata&series=AG.LND.FRST.K2

respectively. By total area burned, however, natural-caused wildfires are the single largest source, accounting for nearly half of the total burned area.[5] Natural-caused wildfires are substantially larger on average than wildfires assigned to human causes.

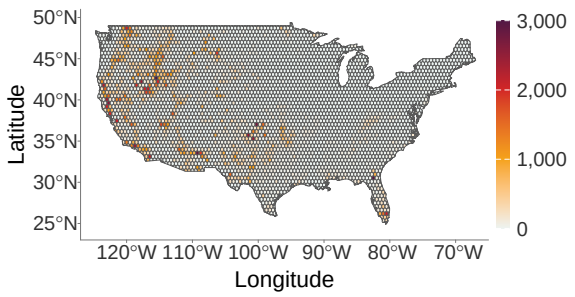
Natural-caused and human-caused wildfires also differ in how predictably they arise. Human-caused wildfires stem from a range of human activities, such as debris and open burning, arson, equipment use, and recreation, and therefore tend to track local economic activity and behavior. Natural-caused wildfires are instead ignited by lightning at largely idiosyncratic locations and times, which makes the large fires they produce difficult to anticipate and adapt to. This randomness in the location and timing of natural-caused wildfires provides a basis for treating their variation as exogenous to the local economic and behavioral conditions that also affect health.

Wildfires occur throughout the US, but burned area is geographically concentrated. Figure 1 compares the distribution of total burned area with the distribution of cumulative HMS smoke-covered area. Panel (a) shows that burned area is concentrated in the western US, with additional areas of substantial burning in Texas and Florida. Panel (b) shows that HMS smoke coverage extends broadly across the central and eastern US. Panel (c) compares the two distributions over the common 2007–2020 period by plotting total burned area and cumulative HMS smoke-covered area within each one-degree longitude band. The largest burned areas occur in western longitude bands, while the largest cumulative smoke-covered areas occur farther east, in the central US. This spatial contrast illustrates why local wildfire size and HMS smoke coverage capture different dimensions of wildfire exposure. Smoke can travel far beyond the locations of the fires that produce it.

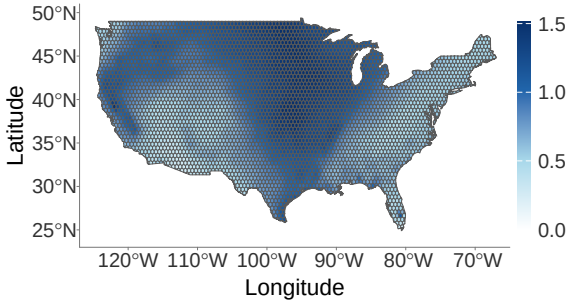
2.2 Wildfires and Lightning Strikes

Lightning strikes are a central ignition source for natural-caused wildfires, and the timing of lightning strikes is associated with the timing of natural-caused wildfires. Figure 2 describes the temporal relationship between wildfires and lightning strikes in the contiguous US over the 2003–2020 analysis period. Panel A plots month-by-year aggregates of natural-caused wildfire size and lightning strikes, showing wildfire size in logs and lightning strikes in levels. The two series are positively correlated, and the natural-caused wildfire size series exhibits a pronounced seasonal pattern that tracks lightning activity. Panel B removes the year dimension by averaging each series by calendar month and compares the seasonal patterns of natural-caused and identified human-caused wildfires. The seasonal pattern is more pronounced for natural-caused wildfires than for identified human-caused wildfires, and the natural-caused series rises sharply in months with higher

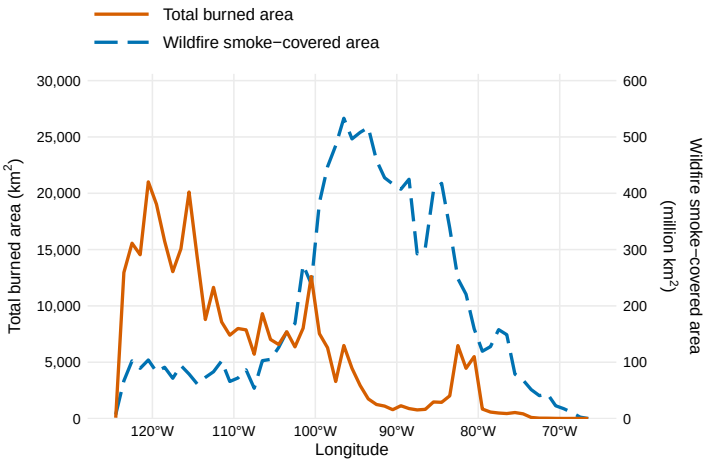
[5]The Fire Program Analysis fire-occurrence database (FPA-FOD) follows the National Wildfire Coordinating Group (NWCG) cause standard. Its public-use records identify natural-caused wildfires but do not distinguish among specific natural ignition causes.



(a) Total Burned Area (km^2)



(b) Wildfire Smoke-Covered Area (million km^2)



(c) Total Areas within One-Degree Longitude Bands

Figure 1: Spatial Distribution and East–West Concentration of Wildfire Burned Area and Wildfire Smoke

Note: Panels (a) and (b) use a regular hexagonal grid with 0.5 degrees between opposite edges. Hexagonal cells along the coast are clipped to the contiguous-US mainland coastline for display. Detached islands are excluded. Panel (a) shows total burned area (km^2) over 2003–2020. Panel (b) shows cumulative HMS smoke-covered area (million km^2) over 2007–2020, summing daily areas and counting repeated coverage on each day. Panel (c) plots totals within one-degree longitude bands over 2007–2020, at band midpoints, with burned area on the left axis and smoke-covered area on the right. Wildfires are assigned by incident longitude and smoke by county centroid. Smoke totals sum daily county exposure areas, which can overlap across counties. HMS records visible smoke plumes rather than ground-level pollution concentrations.

lightning activity.

Dry lightning refers to lightning accompanied by little or no precipitation reaching the ground. When vegetation is dry, the limited rainfall accompanying such lightning leaves fuels susceptible to wildfire ignition. Figure 3 shows the geographic distribution of wildfires and lightning strikes in the contiguous US over the 2003–2020 period. Panel (a) shows the total size of natural-caused wildfires, which are concentrated in the western mountainous region. Panel (b) shows the total number of lightning strikes, while panel (c) presents a proxy for dry lightning, defined as the number of lightning strikes occurring when daily precipitation and specific humidity are below their respective sample medians among hexagon-days with recorded lightning. Lightning strikes overall are more frequent

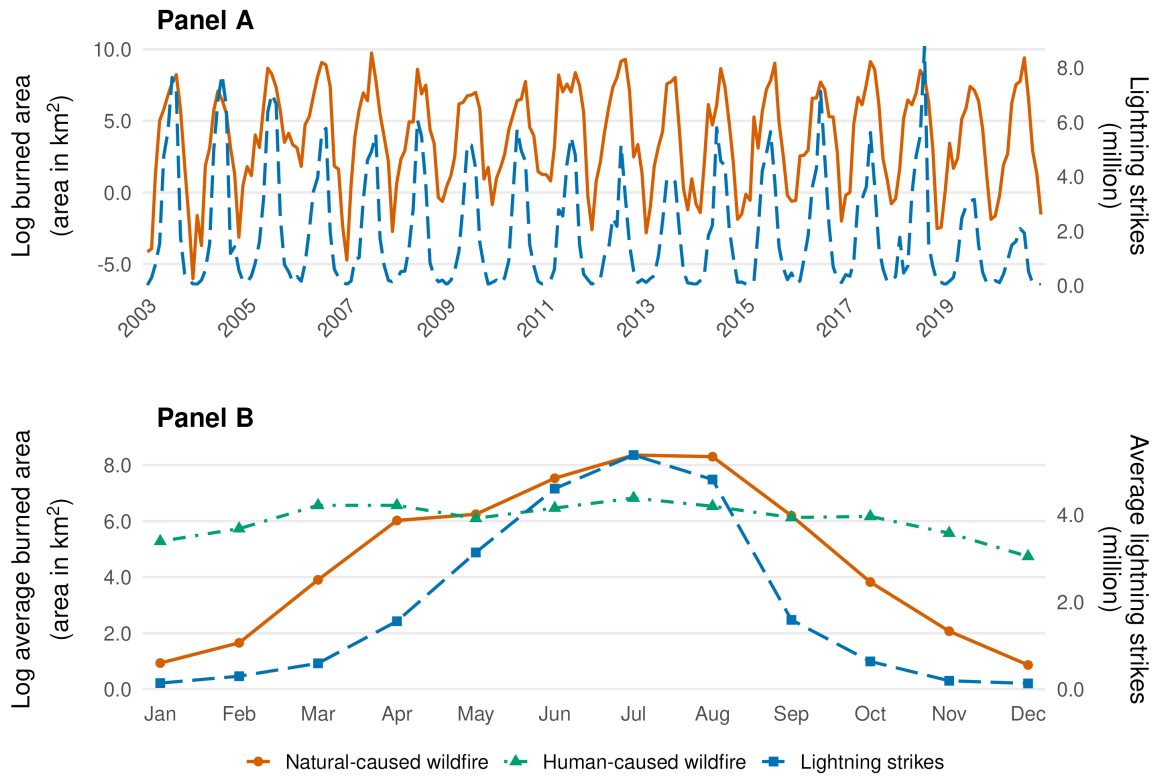


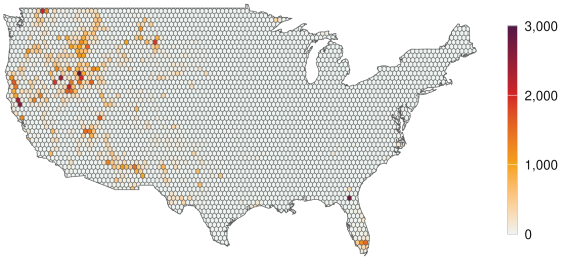
Figure 2: Monthly Patterns in Log Wildfire Burned Area and Lightning Strikes

Note: Panel A plots monthly aggregates for the contiguous US over the 2003–2020 analysis period. The left axis shows the natural logarithm of natural-caused wildfire burned area measured in km^2, while the right axis shows lightning strikes in millions. Panel B averages each series by calendar month over the same period and additionally plots identified human-caused wildfire size, defined as total wildfire size minus natural-caused and undetermined or unclassified wildfire size. The left axis shows the natural logarithm of average natural-caused and identified human-caused wildfire burned area measured in km^2, while the right axis shows average lightning strikes in millions. All wildfire series in the two panels are strictly positive, so no constant is added before taking logarithms.

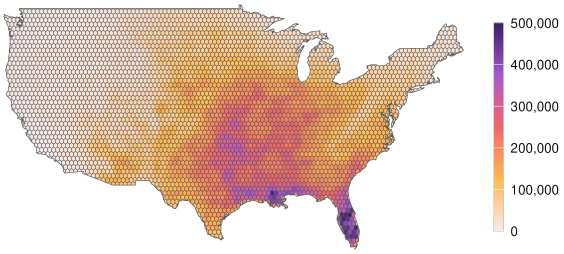
in the southeastern region, including Florida and the Gulf Coast, whereas strikes meeting these low-precipitation and low-humidity criteria are more frequent in the southwestern region, including Arizona and New Mexico. Visually, the spatial distribution in panel (c) more closely resembles that of natural-caused wildfire size in panel (a) than does the distribution of all lightning strikes in panel (b). However, the imperfect spatial correspondence with natural-caused wildfire size may partly reflect the use of a proxy for dry lightning in panel (c).

2.3 Wildfires and Health at Birth

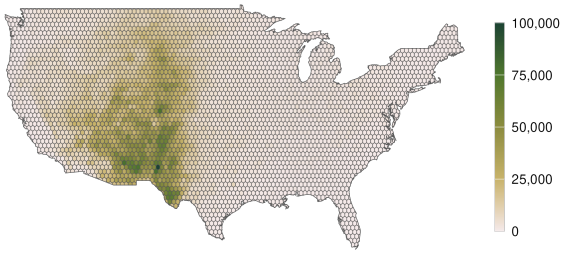
As a large number of studies document, wildfires can affect health at birth through air pollution. Biomass combustion emits particulate matter and other pollutants that damage respiratory and cardiovascular health and are associated with worse birth outcomes (Naeher et al., 2007; Stieb et al., 2012; Currie et al., 2014). Pollution-induced oxidative stress and inflammation can restrict fetal growth by reducing fetal-placental oxygen and nutrient exchange. Accordingly, wildfire-health studies commonly measure exposure



(a) Natural-Caused Wildfire Size (km^2)



(b) Lightning Strikes (count)



(c) Lightning Strikes under Low-Precipitation and Low-Humidity Conditions (count)

Figure 3: Geographic Distribution of Natural-Caused Wildfire Size and Lightning Strikes

Note: All panels summarize observations over 2003–2020. The contiguous-US mainland is divided into a regular hexagonal grid with 0.5 degrees between opposite edges. Detached islands are excluded, while the grid fully covers the mainland polygon and every retained wildfire and lightning location is assigned to one hexagon. Hexagonal cells along the coast are clipped to the contiguous-US mainland coastline for display. Panel (a) shows the total size, measured in square kilometers, of wildfires classified as natural-caused wildfires in the US Forest Service wildfire database. Panel (c) presents a proxy for dry lightning, measured as the number of lightning strikes that occur when daily precipitation and specific humidity are below their respective sample medians among hexagon-days with recorded lightning during 2003–2020.

using HMS smoke plumes from the National Oceanic and Atmospheric Administration (NOAA), satellite-based smoke measures, wildfire-specific $PM_{2.5}$, or wind-driven smoke transport (Fadadu et al., 2021; Borgschulte et al., 2024; Heft-Neal et al., 2022, 2023; Molitor et al., 2023; Cabral and Dillender, 2024; Miller et al., 2024). These measures capture transported smoke, although HMS plume coverage may imperfectly proxy ground-level pollution (Liu et al., 2024). Appendix Table B.1 summarizes the related literature by exposure measure and identification strategy.

Wildfires can also generate evacuation risk, residential exposure, property damage, and uncertainty. These forms of local disruption may affect fetal development through maternal stress-related and behavioral responses. Existing evidence shows that acute prenatal stressors, including terrorist attacks, hurricanes, and earthquakes, worsen birth outcomes (Berkowitz et al., 2003; Camacho, 2008; Brown, 2020; Currie and Rossin-Slater, 2013). In the context of wildfires, recent evidence indicates that wildfire-related mental

health costs operate through smoke $PM_{2.5}$ as well as evacuation orders, direct local costs, and wildfire-related anxiety (Currie and Saberian, 2025). Appendix Table B.2 summarizes related evidence on prenatal stressors and offspring outcomes.

These two channels are not mutually exclusive. Smoke can travel far from the ignition point, while local disruption is more closely tied to local wildfire events. The empirical analysis therefore distinguishes measured smoke exposure from exposure to local wildfire events, then examines suggestive evidence on local disruption, including stress-related responses.

3 Data

3.1 Wildfires

Wildfire data come from the FPA-FOD, compiled by the USFS from the records of multiple federal, state, and local agencies. This study uses records covering 2003–2020 (Short, 2022). This database reports detailed information on wildfire events, such as latitude, longitude, discovery date, final fire size, and fire cause. The county exposure area is defined as the union of the county and a 50 km buffer around its geographic centroid. Wildfire size is constructed by summing the reported final sizes of fires whose recorded locations fall within the county exposure area. Whereas much of the wildfire literature relies on satellite-based fire or smoke products, the FPA-FOD reports a final fire size for each incident, which is better suited to estimating effects along the intensive (size) margin of interest in this study.

This study also uses two wildfire-smoke measures. First, HMS smoke coverage is constructed from NOAA Hazard Mapping System daily smoke-plume shapefiles.[6] Daily HMS smoke coverage is measured as the union area of smoke plumes within the county exposure area, separately for light, medium, heavy, and total smoke. These daily areas are summed within each month to construct smoke-covered area-days for the birth-outcome analysis. Second, wildfire smoke $PM_{2.5}$ is taken from the daily county-level estimates developed by Childs et al. (2022). These estimates isolate wildfire-specific $PM_{2.5}$ from other sources of particulate matter. The analysis uses these county-level estimates directly in county-day pathway analyses and their monthly averages in county-month birth-outcome specifications.

[6]NOAA Hazard Mapping System Fire and Smoke Product: https://www.ospo.noaa.gov/products/land/hms.html. The official daily archive does not supply files for 17 dates in 2006–2020. The construction assigns zero HMS smoke area to those dates. Excluding the 15 affected months does not change the sign or statistical significance of the estimated wildfire-size coefficients.

3.2 Weather and Lightning Strikes

Daily weather data come from the North American Regional Reanalysis (NARR).[7] The NARR combines raw meteorological observations from aircraft, satellites, dropsondes, radiosondes, and land-based weather stations. These data are reported on 32 km × 32 km grids. Weather variables include pressure, temperature, humidity, precipitation, u-wind, and v-wind. The u-wind and v-wind variables are used to construct wind speed.[8] These variables are averaged across NARR grid points within the county exposure area.

Daily counts of cloud-to-ground lightning flashes, referred to here as lightning strikes, come from the National Lightning Detection Network (NLDN) data.[9] The network consists of more than 100 remote ground-based sensing stations located across the US that instantaneously detect the electromagnetic signals emitted when lightning strikes the Earth's surface. This study focuses on cloud-to-ground lightning because it is the primary source of natural-caused wildfire ignitions. Daily counts are reported in 0.1-degree tiles and summed across tiles whose centers fall within the county exposure area, after clipping the area to contiguous US county polygons.

3.3 Birth Outcomes, Mortality, and Maternal Risk Factors

This study uses county-month natality data from CDC WONDER for 2003–2020.[10] The data record births by the mother's legal county of residence and cover 576 counties with populations above 100,000 in the 2000 census. CDC suppresses counts of nine or fewer. The analysis imputes these cells as 5, the midpoint of the possible range, for count-based outcomes. This suppression does not affect the reported county averages of birth weight or gestational age.

The main birth outcomes are county average birth weight and gestational age, low birth weight (below 2,500 grams) per 1,000 births with reported birth weight, and preterm birth (below 37 completed weeks) per 1,000 births with reported gestational age. Birth-outcome regressions are weighted by total births, including births without a reported outcome.

[7] NARR provides precipitation, temperature, humidity, surface pressure, and wind within a common reanalysis framework. Station-based interpolation can be uncertain in complex terrain with sparse observational coverage. The review by Mankin et al. (2025) reports that reanalysis precipitation products can outperform station-based gridded products in mountainous regions with low station density, although performance varies across variables and settings. These considerations motivate using NARR to construct weather controls over the county exposure areas. NARR also incorporates ground observations, and this choice does not imply uniformly greater accuracy than the Parameter-elevation Regressions on Independent Slopes Model (PRISM). NARR data from the National Centers for Environmental Prediction: https://psl.noaa.gov/data/gridded/data.narr.html.

[8] Wind speed $= (u^2 + v^2)^{\frac{1}{2}}$, where v and u are v- and u-wind, respectively.

[9] Severe Weather Database Inventory (SWDI) Lightning Tile Summaries: https://www.ncei.noaa.gov/products/lightning-products

[10] WONDER denotes Wide-ranging Online Data for Epidemiologic Research. CDC WONDER is available at https://wonder.cdc.gov.

Gestational age is measured using last menstrual period (LMP) dating or the obstetric estimate (OE).[11] Although OE generally provides a more accurate measure of gestational age, it is available in these data only from 2007 onward. The main analysis therefore uses LMP to maintain consistent definitions of gestational age and preterm birth over the full 2003–2020 sample. Appendix Table C.1 presents supplementary estimates using OE over 2007–2020, with the comparison described in Appendix C.1.

The analysis also uses maternal age, race and ethnicity, and marital status, together with birth attendant, infant sex, birth order, delivery method, birthplace, and plurality as controls. Maternal-health outcomes from CDC WONDER are pregnancy-associated hypertension, diabetes, and eclampsia. Additional infant-health outcomes are deaths within one day and one year of birth from the CDC WONDER Underlying Cause of Death database.

3.4 Sentiment, Tobacco Use, and Direct-Damage Measures

County-day online sentiment data come from Baylis (2020) and are available for the 2014–2020 portion of the study period.[12] The dataset scores geotagged Twitter posts collected through the Twitter Streaming application programming interface (API) with the Valence Aware Dictionary and sEntiment Reasoner (VADER). County-days without posts remain missing. The analysis standardizes the mean county-day sentiment score using the unweighted mean and standard deviation within the estimation sample. CDC WONDER natality data also provide a county-month rate of reported maternal tobacco use during pregnancy for 2007–2020. The analysis retains county-months with at least 95% reporting coverage.

The analysis uses public data from the US Fire Administration's National Fire Incident Reporting System (NFIRS) on buildings directly involved, property and content losses, and civilian casualties by primary symptom.[13] Reported outcomes are aggregated by month within county boundaries. Because NFIRS reporting is voluntary and incidents are not matched to individual wildfires, these measures capture reported outcomes rather than a complete census of damage from particular fires.

[11] LMP dating measures the interval from the first day of the mother's last normal menstrual period to birth, whereas OE is the birth attendant's final clinical estimate of completed gestational weeks. LMP-based estimates can be affected by errors in maternal recall and irregular menstrual cycles. For definitions and evidence on measurement validity, see the CDC National Center for Health Statistics report *Measuring Gestational Age in Vital Statistics Data: Transitioning to the Obstetric Estimate*: https://www.cdc.gov/nchs/data/nvsr/nvsr64/nvsr64_05.pdf.

[12] Geolocated tweet sentiment data (US county-day, 2014–2022): https://www.patrickbaylis.com/posts/2025-02-22-tweet-data/

[13] NFIRS Public Data Release, US Fire Administration (OpenFEMA): https://www.fema.gov/about/openfema/data-sets/fema-usfa-nfirs-annual-data

3.5 Fuel Treatment

Fuel treatment data come from the hazardous fuel treatment records of the USFS Natural Resource Manager (NRM) Forest Activity Tracking System (FACTS).[14] Because the records primarily cover National Forest System lands, the county measure of fuel-treatment area captures USFS-tracked activity rather than all fuel treatments within a county. The county measures include only the portions of treatment polygons within county boundaries. In the monthly policy regressions, fuel-treatment area sums the areas of mechanical and prescribed-fire treatments completed before the first day of the month, without depreciation in the baseline specification.

Treatments are grouped into prescribed fire, mechanical, chemical, and other categories. Descriptive tables and figures combine chemical and other treatments.[15] This study uses prescribed fire and mechanical treatments as the main treatment types because they are the most common and have the clearest wildfire-mitigation objectives. Appendix E.1 reports area statistics by treatment type and the temporal and geographic distribution of treatment activity (Table E.1 and Figures E.1 and E.2). Table E.1 also reports treatment costs as descriptive background. Table E.2 reports the distribution of beginning-of-month treatment area in the county-month policy estimation sample.

Table 2 reports summary statistics for wildfire size, lightning counts, the main birth outcomes, and the additional health outcomes.

Summary statistics for maternal tobacco use and NFIRS outcomes appear in Appendix A, Table A.1. Monthly smoke-control and daily smoke and sentiment statistics appear in Appendix D, Table D.1.

4 Empirical Strategy

4.1 Identification Issues

Estimating the causal effect of wildfire exposure on birth outcomes is challenging because wildfire activity is not randomly assigned across counties and months. This leads to potential confounding from omitted variables that affect both wildfire exposure and birth outcomes. Debris and open burning, for example, are inexpensive ways to remove crop residues, facilitate planting, and control pests and weeds. These fires may therefore coincide with agricultural production and income gains that can improve household resources and health investments (Rangel and Vogl, 2019; He et al., 2020). A regression using wildfire variation that includes human-caused wildfires could consequently attribute the

[14] USDA Forest Service FSGeodata Clearinghouse (Enterprise Data Warehouse): https://data.fs.usda.gov/geodata/edw/datasets.php

[15] The analysis retains positive-area polygons with a nonmissing completion date, assigns treatment timing using that date, and excludes administrative "Wildfire Non-Treatment" and "Fire Use" records.

Table 2: Summary Statistics for Wildfire Exposure, Lightning, and Health Outcomes

	Obs.	Mean	SD	Min	P10	P90	Max
Panel A. Wildfire and instrument							
Fire size (Natural, km^2)	116,856	0.4	12.6	0.0	0.0	0.0	1,526.0
Fire size (Other, km^2)	116,856	0.9	17.5	0.0	0.0	0.5	1,608.4
No. of Lightning Strikes	116,856	2,271.6	4,525.7	0.0	0.0	7,055.0	73,996.0
Panel B. Birth outcomes							
Ave. Birth Weight (grams)	116,856	3,277.8	76.4	2,861.2	3,180.8	3,373.3	3,604.2
Low Birth Weight per 1,000 Births	116,856	78.5	26.3	0.0	43.5	111.7	225.0
Ave. LMP Gestational Age (weeks)	116,856	38.6	0.3	34.7	38.2	39.0	42.2
Ave. OE Gestational Age (weeks)	91,896	38.5	0.3	36.9	38.1	38.8	39.6
LMP Preterm Births per 1,000 Births	116,856	117.2	35.7	0.0	76.5	158.2	714.3
OE Preterm Births per 1,000 Births	91,896	96.9	28.9	0.0	58.1	132.5	288.6
Panel C. Additional birth and infant outcomes and maternal risk factors							
Live Births per 1,000 Population	116,856	1.1	0.2	0.2	0.8	1.3	2.7
Infant Mortality (1st Day) per 1,000,000 Population	116,856	8.3	11.9	0.0	0.0	28.0	52.4
Infant Mortality (1st Year) per 1,000,000 Population	116,856	14.9	13.6	0.0	0.0	36.0	70.0
Pregnancy-Associated Hypertension per 1,000 Births	116,856	49.4	32.5	0.0	0.0	92.7	281.2
Diabetes per 1,000 Births	116,856	54.5	33.2	0.0	0.0	97.9	355.4
Eclampsia per 1,000 Births	116,856	4.9	10.7	0.0	0.0	20.0	116.3

Notes: This table reports unweighted summary statistics, including the number of nonmissing county-month observations (Obs.), mean, standard deviation (SD), minimum (Min), tenth percentile (P10), ninetieth percentile (P90), and maximum (Max). Panel A reports natural-caused wildfire size, other wildfire size, and lightning counts in the birth-outcome sample. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Wildfire areas are reported before the logarithmic transformation used in the regressions, which adds 10^{-6} km^2 to each measure. Panel B reports the main birth outcomes and alternative obstetric-estimate (OE) measures of gestational age and preterm birth. Panel C reports additional birth and infant outcomes and maternal risk factors. All rows cover 2003–2020 except the OE rows in Panel B, which cover 2007–2020. Average birth weight is measured in grams, and gestational age is measured in weeks. Low birth weight and preterm birth are measured per 1,000 births with the corresponding outcome reported, live births per 1,000 population, infant mortality per 1,000,000 population, and maternal risk factors per 1,000 births. CDC-suppressed count cells of nine or fewer are imputed as 5 when constructing count-based rates.

effects of these economic conditions to wildfire exposure.

Other human-caused wildfires, including arson, may also reflect local socioeconomic conditions. Human-caused wildfire ignitions in the US occur in more densely populated areas than natural-caused ignitions (Pourmohamad et al., 2024). For arson, an additional concern is that criminal activity is related to local labor market opportunities and regional income (Gould et al., 2002), as well as urban conditions and financial returns to crime (Glaeser and Sacerdote, 1999). Changes in these socioeconomic conditions may also affect birth outcomes through household resources and access to health care. Although county fixed effects absorb persistent differences across locations, changes in local conditions may jointly influence human-caused wildfire activity and birth outcomes. This concern motivates isolating lightning-induced variation in natural-caused wildfire size.

4.2 Instrumental Variable Estimation

To address the endogeneity of human-caused wildfires discussed above, this study constructs natural-caused wildfire size from the USFS cause classification and uses it as the main treatment variable.[16] This restriction reduces concerns that wildfire exposure is driven by local economic activity, agricultural burning, arson, recreation, or other human

[16]The natural-caused category follows the NWCG cause standard, but the public-use records do not distinguish among specific natural ignition causes, as described in Section 2.

behaviors that may also affect birth outcomes. Lightning strikes serve as the instrument for this cause-classified measure.

This study estimates the effect of natural-caused wildfires on birth outcomes at the county-month level. The baseline specification takes the following form.

$$Y_{ct} = \beta \ln(NaturalFire_{ct}) + X'_{ct}\gamma + \alpha_c + \lambda_t + \varepsilon_{ct}, \tag{1}$$

where the dependent variable is the outcome of interest in county c in month t (Y_{ct}). Birth outcomes are county-month averages of birth weight and gestational age, as well as the county-month rates of low birth weight and preterm birth. The main variable of interest, $\ln(NaturalFire_{ct})$, is the logarithm of natural-caused wildfire size aggregated over the county exposure area in the birth month. Because the logarithm is undefined at zero, the baseline and related specifications add 10^{-6} to each wildfire-size variable before taking logs. This additive constant is suppressed in the displayed notation. Appendix C.3 considers exploratory specifications with lagged wildfire-size terms, but the baseline specification focuses on the birth-month measure because timing effects are difficult to interpret when outcomes are indexed by realized birth month.

The vector X_{ct} includes logged fire weather conditions (i.e., log humidity, log precipitation, log air temperature, and all possible interactions among them) and log other wildfire size. Other wildfire size equals total wildfire size minus natural-caused wildfire size and therefore combines identified human-caused wildfires with wildfires of undetermined or unclassified causes.[17] The county fixed effects (α_c) control for time-invariant confounders specific to a county, such as forest area, soil condition, and geographic differences in health and pollution. The year-month fixed effects (λ_t) control for common time-varying shocks, such as changes in medical or environmental policies for wildfire management during the sample period.

The OLS estimates of Equation 1 use the observed cause-classified treatment variable directly. However, a limitation of this approach is that the cause classification remains incomplete. Table 1 shows that 29.1% of wildfire incidents are not classified by cause. Some lightning-related fires may be missing from the natural-caused category, so the observed measure of natural-caused wildfire size may contain measurement error, which may bias the OLS estimates. Therefore, this study also estimates the causal effect of natural-caused wildfires using an IV approach. The first-stage specification takes the following form.

[17]The report by National Wildfire Coordinating Group (2023) states that the four critical weather elements that produce extreme fires are low relative humidity, strong surface winds, unstable air, and drought.

$$\ln(NaturalFire_{ct}) = \pi Lightning_{ct} + X'_{ct}\kappa + \mu_c + \tau_t + \nu_{ct}, \tag{2}$$

where $Lightning_{ct}$ is the lightning count for county c's exposure area in month t, constructed as described in Section 3.

The IV approach relies on relevance and the exclusion restriction. For the baseline specifications, the outcome-specific effective first-stage F-statistics of Montiel Olea and Pflueger (2013) reported in Table 5 indicate a strong first-stage relationship. The exclusion restriction requires that lightning strikes affect birth outcomes only through their effect on natural-caused wildfire size. Table 1 shows that natural-caused wildfires are larger on average than human-caused wildfires. This pattern is consistent with evidence that natural-caused ignitions tend to occur in less densely populated areas and on higher and steeper terrain (Pourmohamad et al., 2024). Therefore, lightning strikes are unlikely to be correlated with local economic activity, agricultural burning, arson, or other human behaviors that may also affect birth outcomes.

A concern is that lightning strikes may affect birth outcomes through storm-related disruptions. For example, lightning strikes are associated with thunderstorms, which can cause property damage and other disruptions that may affect birth outcomes.[18] To address these concerns, additional specifications control for storm events and their property damage as a robustness check (Appendix Table C.5). The main results are robust to these controls. Appendix Figure C.2 shows markedly different geographic distributions for the dry-lightning proxy and reported storm activity. Therefore, the identifying variation in lightning strikes is unlikely to be driven by storm-related disruptions.

5 Results

5.1 Lightning Strikes and Wildfires

Table 3 reports the effect of lightning strikes on natural-caused wildfires at the county-month level, using specifications with county and year-month fixed effects. Columns 1 to 3 use log wildfire size as the dependent variable, while Columns 4 to 6 use the number of wildfires.

The baseline estimates in Columns 1 and 4 show that lightning strikes increase both burned area and the number of natural-caused wildfires. Columns 2 and 5 add controls for atmospheric pressure and wind speed to the baseline specifications. The coefficients on lightning strikes change little, indicating that the results are robust to these additional

[18] Zou et al. (2020) find that thunderstorms increase the number of emergency visits related to respiratory illness.

weather controls.

Columns 3 and 6 allow the effect of lightning strikes to vary with temperature, precipitation, and humidity. The positive temperature interactions and negative humidity interactions suggest that both burned area and fire counts respond more strongly to lightning strikes under hotter and drier conditions. The precipitation interactions are also negative, although the estimate for fire counts is less precise.[19]

Table 3: Lightning Strikes and Wildfires (OLS, county-month level), 2003–2020

	Natural-Caused Wildfire					
	Ln Size			Count		
	(1)	(2)	(3)	(4)	(5)	(6)
Std. No. of Lightning Strikes	0.581***	0.591***	0.407**	0.835***	0.850***	0.920**
	(0.086)	(0.084)	(0.161)	(0.210)	(0.208)	(0.377)
Std. No. of Lightning Strikes × Std. Temp.			0.929***			2.338***
			(0.245)			(0.856)
Std. No. of Lightning Strikes × Std. Precip.			-0.564***			-0.295
			(0.139)			(0.223)
Std. No. of Lightning Strikes × Std. Humidity			-0.423*			-2.288**
			(0.253)			(1.045)
County FE	X	X	X	X	X	X
Year-Month FE	X	X	X	X	X	X
Fire Weather Controls	X	X	X	X	X	X
Other Weather Controls		X	X		X	X
Mean (Dep. Var.)	0.440	0.440	0.440	0.869	0.869	0.869
SD (Dep. Var.)	12.635	12.635	12.635	4.672	4.672	4.672
Counties	576	576	576	576	576	576
Obs.	116,856	116,856	116,856	116,856	116,856	116,856
R^2	0.453	0.457	0.469	0.281	0.287	0.332

Notes: The dependent variable is log natural-caused wildfire size in Columns 1 to 3 and the number of natural-caused wildfires in Columns 4 to 6. The observational unit is the county-month in the birth-outcome sample. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, reporting wildfire size before the log transformation in Columns 1 to 3 and wildfire counts in Columns 4 to 6. Lightning counts and the fire weather variables labeled "Std." are standardized using unweighted means and standard deviations calculated within this estimation sample. Fire weather controls include log precipitation, log humidity, log air temperature, and all interactions among these variables. All columns also control for log other wildfire size, where other wildfire size equals total wildfire size minus natural-caused wildfire size and therefore combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Other weather controls are pressure and wind speed. All columns include county fixed effects and year-month fixed effects, are weighted by the number of births, and report standard errors clustered by county in parentheses. Significance stars follow the conventional notation.

5.2 Wildfires and Health

OLS Results by Cause of Wildfires.

Next, this study examines the county-month relationship between wildfire size and birth outcomes. Before turning to the IV estimates, Table 4 reports OLS associations disaggregated by wildfire ignition cause. Each specification includes log natural-caused

[19] A first stage that varies with covariates can complicate the interpretation of conventional two-stage least squares, which enters the instrument additively and thereby restricts its effect to be homogeneous across covariate values (Słoczyński, 2020). As a check, Appendix Table C.2 re-estimates the model with the lightning instrument interacted with the fire weather covariates following Angrist and Imbens (1995), as described in Appendix C.2. The main conclusions are unchanged.

wildfire size, log debris and open burning fire size, log arson fire size, and log remaining fire size, which combines the other ten cause categories. Columns 1 to 4 use average birth weight, low birth weight (below 2,500 grams), average gestational age, and preterm birth (below 37 weeks) as dependent variables, respectively.

The coefficient on natural-caused wildfire size is adverse across all four outcomes. For example, a one standard deviation increase in natural-caused wildfire size from its sample mean is associated with birth weight lower by 0.4 grams and gestational age shorter by 0.001 weeks, with standard errors small relative to these estimates. In addition, the same increase is associated with higher rates of low birth weight and preterm birth by 0.1 per 1,000 births and 0.2 per 1,000 births, respectively.

The mixed OLS associations for human-caused wildfires are consistent with the identification concerns in Section 4.1. These fires may track agricultural activity or socioeconomic conditions that also affect birth outcomes. The positive association between arson-related wildfire size and birth weight is therefore unlikely to represent a causal health benefit. These patterns motivate the IV strategy, which isolates lightning-induced variation in natural-caused wildfire size.

Table 4: Wildfires and Birth Outcomes by Ignition Cause (OLS, county-month level), 2003–2020

	Birth Weight (grams)	Low Birth Weight per 1,000 Births	LMP Gestational Age (weeks)	Preterm Births per 1,000 Births
	(1)	(2)	(3)	(4)
Ln Natural-Caused Fire Size	-0.1207***	0.0229*	-0.0004**	0.0572**
	(0.0366)	(0.0134)	(0.0002)	(0.0242)
Ln Debris/Open-Burning Fire Size	0.0030	-0.0209	0.0003	-0.0718**
	(0.0407)	(0.0136)	(0.0003)	(0.0351)
Ln Arson/Incendiarism Fire Size	0.0812**	-0.0242	-0.0002	0.0251
	(0.0410)	(0.0151)	(0.0003)	(0.0346)
Ln Remaining Fire Size	0.0137	0.0096	-0.0002	-0.0374
	(0.0463)	(0.0166)	(0.0003)	(0.0283)
County FE	X	X	X	X
Year-Month FE	X	X	X	X
Fire Weather Controls	X	X	X	X
Other Weather Controls	X	X	X	X
Mean (Dep. Var.)	3278	79	38.6	117
SD (Dep. Var.)	76	26	0.3	36
Counties	576	576	576	576
Obs.	116,856	116,856	116,856	116,856
R^2	0.767	0.475	0.587	0.501

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. All estimations include county fixed effects, year-month fixed effects, fire weather controls, pressure, and wind speed. Estimations are weighted by the total number of births, including births with missing outcome-reporting status. Standard errors clustered by county are shown in parentheses. Each fire-size variable is the logarithm of wildfire size in the birth month for that cause category. Remaining Fire Size is total wildfire size minus natural-caused wildfire size, debris and open burning fire size, and arson/incendiarism fire size. It combines the other ten cause categories in FPA-FOD. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

IV Results.

Table 5 compares OLS and IV estimates for the four birth outcomes in the county-month panel. The IV estimates use lightning strikes to isolate variation in natural-caused wildfire size generated by plausibly exogenous ignition pressure. The IV point estimates

Table 5: Wildfires and Birth Outcomes (OLS and IV, county-month level), 2003–2020

	Birth Weight (grams)		Low Birth Weight per 1,000 Births		LMP Gestational Age (weeks)		Preterm Births per 1,000 Births	
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)	OLS (7)	IV (8)
Ln Natural-Caused Fire Size	-0.118***	-1.219***	0.022	0.311**	-0.0005**	-0.0035*	0.058**	0.272
	(0.037)	(0.334)	(0.013)	(0.122)	(0.0002)	(0.0018)	(0.025)	(0.197)
County FE	X	X	X	X	X	X	X	X
Year-Month FE	X	X	X	X	X	X	X	X
Weather Controls	X	X	X	X	X	X	X	X
F^{Eff}		49.97		49.97		49.97		49.97
Mean (Dep. Var.)	3278	3278	79	79	38.6	38.6	117	117
SD (Dep. Var.)	76	76	26	26	0.3	0.3	36	36
Counties	576	576	576	576	576	576	576	576
Obs.	116,856	116,856	116,856	116,856	116,856	116,856	116,856	116,856
R^2	0.767	0.764	0.475	0.474	0.587	0.586	0.501	0.500

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. Columns 1, 3, 5, and 7 report OLS estimates. Columns 2, 4, 6, and 8 report IV estimates that instrument log natural-caused wildfire size with lightning counts. All estimations include log other wildfire size, county fixed effects, year-month fixed effects, fire weather controls, pressure, and wind speed. Estimations are weighted by the total number of births, including births with missing outcome-reporting status. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. F^{Eff} is the outcome-specific effective first-stage F-statistic for the IV specifications. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

have adverse signs across all four birth outcomes. A one standard deviation increase in natural-caused wildfire size from its sample mean reduces birth weight by 4.1 grams and increases the incidence of low birth weight by 1.1 per 1,000 births.[20]

The IV point estimates indicate shorter gestation and more preterm births, although their magnitudes remain uncertain. Using the LMP definition, the same increase reduces gestational age by 0.01 weeks and increases the incidence of preterm birth by 0.9 per 1,000 births. Appendix C.1 compares the two definitions over 2007–2020. The OE estimates retain the adverse signs and have smaller standard errors. This comparison is consistent with LMP measurement error contributing to uncertainty. Across all four outcomes, these IV estimates are larger in magnitude than their OLS counterparts. The gap may reflect OLS bias toward zero or different weighting of heterogeneous effects across county-month conditions and wildfire sizes (Ishimaru, 2024). The present comparison does not distinguish these explanations.

Exposure Timing.

Appendix Table C.3 reports an exploratory distributed-lag IV specification with wildfire exposure in the birth month and three and six months before birth, instrumented by lightning counts in the corresponding months. Because the lags are indexed to the realized birth month and wildfire exposure may affect gestational length, they do not map cleanly into trimesters. Adding longer lags weakens identification.

[20]Since the natural-caused wildfire-size variable is in logarithms, a one standard deviation increase from the sample mean corresponds to changing wildfire size from μ_F to $\mu_F + \sigma_F$, where σ_F and μ_F are the standard deviation and mean of natural-caused wildfire size, respectively. The reported effect is $\beta \ln(1 + \sigma_F/\mu_F)$.

Birth-weight estimates are negative at all three lags, and gestational-age estimates are also negative, with the clearest response three months before birth. The remaining outcomes show less consistent patterns, precluding a sharper trimester-specific interpretation.

Appendix Table C.4 complements this analysis with estimates based on average exposure over expanding windows. Birth-weight and gestational-age estimates retain adverse signs through the better-identified windows, but the effective F-statistic falls below 10 beyond six months and standard errors widen. These timing estimates should therefore be interpreted cautiously, and the main analysis focuses on the birth-month exposure measure.

Dose Response.

Appendix Figure C.1 provides a descriptive check of the functional form underlying the baseline log-linear specification. The plots replace log natural-caused wildfire size with mutually exclusive quartile indicators for positive fire-size months, using county-months with no natural-caused wildfire as the reference category. Each point is placed at the within-bin mean of log positive fire size, and the histogram at the base of each plot shows the distribution of positive fire-size observations. These estimates are not IV estimates. They are fixed-effects, covariate-adjusted diagnostics for the shape of the conditional relationship.

The figure shows that the relationship between wildfire size and birth outcomes is generally adverse, with larger wildfires associated with lower birth weight, shorter gestation, and higher rates of low birth weight and preterm birth. The relationship is approximately linear in log wildfire size, supporting the baseline log-linear specification, especially for average birth weight. However, the gradients are not monotonic for all outcomes and the confidence intervals are wide.

Mortality and Maternal Risk Factors.

Table 6 reports the effect of wildfires on mortality and maternal risk factors. Column 1 uses the number of live births per 1,000 population as a broader proxy for in utero survival, a strategy similar to that used in Jayachandran (2009) and Rangel and Vogl (2019). A one standard deviation increase in natural-caused wildfire size from its sample mean reduces the number of live births by approximately 0.03 per 1,000 population.

For postnatal mortality, Columns 2 and 3 use deaths occurring in the observation month among infants aged under one day and under one year, respectively, per 1,000,000 population. The analysis matches these deaths to wildfire exposure in the month of death. It does not follow a birth cohort over its first year of life. The point estimates imply that a one standard deviation increase in natural-caused wildfire size from its sample mean raises deaths among infants aged under one day by 0.01 per 1,000,000 population and among

infants aged under one year by 0.3 per 1,000,000 population, although the standard errors are large.

For maternal risk factors, Columns 4 to 6 use reports of pregnancy-associated hypertension, diabetes, and eclampsia per 1,000 births as outcomes. These outcomes help assess whether wildfire exposure affects maternal health directly and whether maternal-health responses may contribute to the birth-outcome effects. The point estimates imply a decrease in pregnancy-associated hypertension of 0.4 per 1,000 births and an eclampsia effect close to zero, with an absolute magnitude of 0.01 per 1,000 births. Both estimates are small relative to their standard errors, leaving the direction and magnitude of these maternal-health responses uncertain.

Table 6: Wildfires and Additional Health Outcomes (IV, county-month level), 2003–2020

	Births	Infant Mortality		Maternal Risk Factors		
	Live Births per 1,000 Population	Deaths under 1 day per 1,000,000 Population	Deaths under age 1 per 1,000,000 Population	Hypertension per 1,000 Births	Diabetes per 1,000 Births	Eclampsia per 1,000 Births
	(1)	(2)	(3)	(4)	(5)	(6)
Ln Natural-Caused Fire Size	-0.009***	0.003	0.077*	-0.120	0.832***	-0.004
	(0.002)	(0.042)	(0.046)	(0.218)	(0.220)	(0.048)
County FE	X	X	X	X	X	X
Year-Month FE	X	X	X	X	X	X
Fire Weather Controls	X	X	X	X	X	X
Other Weather Controls	X	X	X	X	X	X
F^{Eff}	49.97	49.97	49.97	49.97	49.97	49.97
Mean (Dep. Var.)	1.1	8.3	14.9	49.4	54.5	4.9
SD (Dep. Var.)	0.2	11.9	13.6	32.5	33.2	10.7
Counties	576	576	576	576	576	576
Obs.	116,856	116,856	116,856	116,856	116,856	116,856
R^2	0.858	0.185	0.392	0.601	0.596	0.213

Notes: The dependent variables are live births per 1,000 population, deaths among infants aged under one day and under one year per 1,000,000 population, and reports of pregnancy-associated hypertension, diabetes, and eclampsia per 1,000 births. The observational unit is the county-month. Mortality counts are grouped by month of death and matched to wildfire exposure in that month. The mortality denominator is total county population, and these outcomes do not measure mortality in a birth cohort followed over time. All columns report IV estimates of log natural-caused wildfire size instrumented by lightning counts. All estimations include log other wildfire size, fire weather controls, pressure, wind speed, county fixed effects, and year-month fixed effects, and are weighted by the number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics within each outcome estimation sample. Significance stars follow the conventional notation.

In contrast, a one standard deviation increase in natural-caused wildfire size from its sample mean raises reported maternal diabetes by 2.8 per 1,000 births, with a standard error small relative to the estimate. Potential pathways are reduced physical activity during smoke exposure and local disruption (McIntyre et al., 2019; Chen et al., 2024) and pollution-induced inflammation and oxidative stress (Yang et al., 2020). These channels remain suggestive because exercise is unobserved, the measure may combine gestational and pre-existing cases, and gestational diabetes can raise birth weight (McIntyre et al., 2019).

Robustness checks.

Appendix Table C.5 reports specifications with additional controls and fixed effects. Column 1 is the baseline specification. Column 2 adds infant and birth characteristics, including infant sex, birth attendant, place of birth, and plurality. Column 3 adds maternal characteristics, including age, birth order, Hispanic origin, race, and marital status, to assess sensitivity to observed maternal composition. These controls yield birth-weight and low-birth-weight estimates close to the baseline, reducing concern that changes in observed maternal composition explain the main results.

Column 4 controls for storm events and property damage because lightning may be correlated with thunderstorms that directly affect birth outcomes (Zou et al., 2020). The storm controls do not change the main estimates. Appendix Figure C.2 also shows that lightning strikes under low-precipitation and low-humidity conditions have a different spatial distribution from storm events and property damage. Together, these checks reduce, but do not eliminate, concern about storm-related violations of the exclusion restriction. Columns 5–7 allow for county-specific seasonality, one-month-lagged weather, and county-by-year fixed effects, respectively, to control for potential confounding from seasonal and time-varying county characteristics. Across these specifications, the estimates retain their baseline signs, although allowing for county-specific seasonality increases uncertainty and county-by-year fixed effects change some magnitudes.

Power outages provide another potential pathway through which lightning could affect birth outcomes.[21] To address this concern, this study uses county-level power-outage data from the Environment for Analysis of Geo-Located Energy Information (EAGLE-I) system maintained by Oak Ridge National Laboratory, which records the number of customers without power at fifteen-minute intervals. Since the EAGLE-I data are available from November 2014 onward, the analysis is limited to the November 2014–December 2020 subsample. Appendix Table C.6 reports estimates with county-level power-outage controls. The wildfire estimates remain similar to the baseline.

A separate concern is the treatment of county-months with no wildfire. The baseline specification adds a small constant to wildfire size before taking logs, which imposes a particular scaling of changes from zero to positive wildfire size (Chen and Roth, 2024). To assess sensitivity to this choice, Appendix Table C.7 reports alternative specifications that normalize positive natural-caused wildfire size by its minimum positive value before taking logs and assign zero-fire observations values of 0, -0.1, -1, -3, or -5. Because the smallest positive wildfire size maps to zero after this normalization, these assignments set the gap between no wildfire and the smallest positive wildfire size to 0, 0.1, 1, 3, or 5 log units, respectively. Across these specifications, the estimates retain their baseline signs and remain broadly similar in magnitude.

Appendix Tables C.8 and C.9 also assess sensitivity to spatial correlation using Conley standard errors across alternative distance cutoffs (Conley, 1999). The resulting uncertainty estimates do not alter the interpretation of the baseline results.

Finally, Appendix Table C.10 instruments natural-caused wildfire size one to three months after birth with the corresponding lightning counts. These future exposures should not affect in utero health, although the first post-birth month may partly capture fires that continue beyond the birth month. The post-birth estimates are small, and the joint tests do not reveal a systematic placebo pattern after accounting for the four

[21] Burlando (2014b) finds that prenatal exposure to a month-long blackout in Zanzibar reduced birth weight and increased the incidence of low birth weight.

outcomes.

5.3 Wildfire Smoke and Local Disruption

A large literature studies the health and economic effects of wildfire smoke using NOAA HMS smoke plume data (e.g., Fadadu et al., 2021; Borgschulte et al., 2024; Heft-Neal et al., 2022; Molitor et al., 2023; Cabral and Dillender, 2024; Miller et al., 2024). HMS measures the geographic footprint of visible smoke plumes, which can extend far downwind from the fires that produce them. The IV design in this study uses lightning-induced variation in natural-caused wildfire size to capture exposure to local wildfire events and their consequences. The IV estimates indicate that larger natural-caused wildfires increase both HMS smoke coverage and wildfire-specific $PM_{2.5}$ concentrations (Appendix Table D.2).[22] Figure 1 also shows that burned area and HMS smoke coverage have different spatial distributions. Total burned area is concentrated in the western US, whereas HMS smoke coverage extends broadly across the central and eastern US.

Local wildfires can also expose residents to physical danger, material losses, and disruption of daily life. These consequences may generate psychological and behavioral responses, while smoke exposure itself may also affect mental health (Currie and Saberian, 2025). The remaining analyses compare birth-outcome estimates with and without smoke controls and examine Twitter sentiment, maternal tobacco use, reported fire damage, and demographic heterogeneity in birth outcomes. These analyses document several consequences of local wildfire events but do not identify their relative contributions to the birth-outcome effects.

Smoke Controls.

Appendix Table D.3 adds smoke-exposure measures to the baseline birth-outcome IV specification. Column 1 reports the baseline estimate for the sample with available smoke measures. Columns 2 and 3 add HMS smoke area and the wildfire smoke $PM_{2.5}$ estimates developed by Childs et al. (2022), respectively. Because smoke exposure is a consequence of wildfire events, these specifications provide a descriptive comparison of the fire-size estimates with and without smoke controls. The coefficients on both smoke measures describe conditional associations with birth outcomes.

The coefficient on natural-caused wildfire size remains similar across these specifications. HMS smoke area is associated with higher birth weight, lower low-birth-weight incidence, and longer gestation. Its association with preterm birth is negative but impre-

[22] The estimates use a common county-month sample for 2006–2020, with HMS exposure measured as the monthly sum of daily smoke-covered area and wildfire-specific $PM_{2.5}$ as the monthly average concentration. A one standard deviation increase in monthly natural-caused wildfire size from its sample mean is estimated to raise HMS smoke area-days by 10,106 km^2-days and wildfire-specific $PM_{2.5}$ by 0.40 $\mu g/m^3$.

cisely estimated. The wildfire smoke $PM_{2.5}$ coefficients indicate lower birth weight, lower low-birth-weight incidence, shorter gestation, and more preterm births, with substantial uncertainty around several estimates. Thus, controlling for the available smoke measures does not materially alter the estimated birth-outcome effects of wildfire size. However, if these smoke measures contain measurement error, the stability of the wildfire-size coefficients does not rule out effects operating through air pollution (Pei et al., 2019).

Sentiment and Maternal Tobacco Use.

The analysis next examines public sentiment and maternal tobacco use as additional outcomes of wildfire exposure. Twitter sentiment captures short-run changes in public mood among users posting geolocated tweets, while maternal tobacco use records a behavior with potential implications for infant health.[23]

Appendix Table D.4 reports county-day IV estimates using standardized Twitter sentiment as the outcome. A one standard deviation increase in natural-caused wildfire size from its sample mean lowers sentiment by 0.44 standard deviations, consistent with an adverse change in public mood following local wildfire events. This decline is similar in standardized magnitude to the reduction in daily sentiment following nearby hurricanes reported by Baylis (2020).[24]

Turning to behavioral responses, Appendix Table D.5 reports an increase in maternal tobacco use. A one standard deviation increase in natural-caused wildfire size from its sample mean raises maternal tobacco use by 4.388 per 1,000 births with a reported tobacco status. Applying the adjusted smoking estimate in Almond et al. (2005) to this increase implies an illustrative reduction in average birth weight of 0.89 grams. Evaluated at the same change in log wildfire size, the implied reduction is approximately 21% of the baseline birth-weight effect.[25] The increase in tobacco use is consistent with stress-related coping, but the analysis does not distinguish this explanation from other behavioral responses to wildfires.

Direct Damage.

This study also examines county-month fire outcomes reported in NFIRS to assess

[23]Psychological stress predicts smoking initiation and relapse (Kassel et al., 2003), and disaster-related psychological distress has been linked to increased tobacco consumption (Reijneveld et al., 2003; Vlahov et al., 2002; Parslow and Jorm, 2006; Nakano et al., 2018; Osaki et al., 2020). Prenatal tobacco use is also a well-documented risk factor for adverse birth outcomes.

[24]Baylis (2020) reports that hurricanes reduce sentiment by approximately 0.4 standard deviations. The comparison provides a benchmark for magnitude, although the estimates use different geographic units and standardization samples.

[25]The calculation uses the adjusted birth-weight difference of 203.2 grams reported in Table VI of Almond et al. (2005). The percentage compares the implied birth-weight coefficient with the baseline IV coefficient. It assumes that the external smoking estimate applies to this setting. The calculation uses different estimation samples and does not estimate the individual-level relationship between wildfire-induced changes in smoking and birth outcomes. The tobacco reports do not date changes in smoking relative to wildfire exposure. It is a benchmark for magnitude rather than an identified mediation share.

forms of local damage that HMS smoke coverage does not measure directly. Buildings involved and civilian casualties indicate physical danger, while property and content losses per resident measure material damage and potential economic disruption. NFIRS incidents are not individually matched to the wildfires used to construct county-month wildfire size, so the analysis links these outcomes to wildfire exposure at the county-month level.

Appendix Table D.7 shows that larger natural-caused wildfires increase the reported number of buildings involved and property and content losses per resident. A one standard deviation increase in natural-caused wildfire size from its sample mean raises the reported number of buildings involved by 0.45 per 100,000 residents. The same increase raises one plus per-resident property losses by 7.1% and one plus per-resident content losses by 4.6%.[26] These results document physical exposure and material losses associated with local wildfire events.

Appendix Table D.8 reports positive estimates for total civilian casualties and for respiratory- and burn-related casualties. The respiratory- and burn-specific estimates are uncertain, providing limited evidence on the types of injury associated with wildfire exposure. The casualty estimates are consistent with physical risks associated with local wildfire events.

Heterogeneous Effects.

Finally, Appendix Table D.6 examines whether the birth-outcome effects of wildfires vary with county-month maternal demographic profiles. The specification interacts wildfire size with standardized measures of the Black maternal share, unmarried maternal share, and average maternal age. Each demographic measure is standardized to mean zero and unit standard deviation without weights within the common estimation sample. These interactions are instrumented with the corresponding interactions between lightning counts and the demographic measures. The Sanderson–Windmeijer F-statistics support instrument relevance in these specifications.

The estimates indicate more adverse effects on birth weight, low birth weight, and gestational age in county-months with higher Black maternal shares.[27] The preterm-birth interaction is small and uncertain. The point estimates also suggest more adverse effects on low birth weight in county-months with higher unmarried maternal shares, and on birth weight and low birth weight in those with younger maternal-age profiles.

The patterns for unmarried maternal share and maternal age indicate differences

[26] For each loss outcome L, measured in dollars per resident, the percentage change in $1+L$ is calculated as $100[\exp(\Delta) - 1]$, where Δ is the estimated change in $\ln(1 + L)$.

[27] Evidence on racial heterogeneity in the pollution literature is mixed, with some studies finding larger effects among Black mothers and others finding limited or contrasting differences (Currie and Walker, 2011; Heft-Neal et al., 2022; Alexander and Schwandt, 2022). The race interaction may reflect differences in exposure, baseline health, socioeconomic conditions, health-care access, or avoidance capacity, and does not distinguish among these pathways.

in vulnerability to wildfire exposure. Studies of disasters report greater adverse effects among single and younger mothers (Kim et al., 2017; de Oliveira et al., 2023). Such differences may reflect exposure, baseline health, household resources, health-care access, or behavioral responses. However, the heterogeneity estimates describe differences across county-month demographic profiles and cannot be interpreted as individual-level differences in treatment effects. They therefore do not distinguish among the pathways linking wildfires to birth outcomes.

6 Policy Implication: Fuel Treatment

6.1 Fuel Treatment and Wildfire Damage

Fuel treatment is a primary policy tool for managing wildfire risk. The two main types of fuel treatment are mechanical removal and prescribed fire. The mechanical removal of fuel involves the physical removal of vegetation, such as trees, shrubs, and grasses, to reduce the amount of combustible material in an area. Prescribed fire, also known as controlled burning, is the intentional use of fire under controlled conditions. The US Forest Service and partner agencies treat several million acres of public land each year.

Recent evidence indicates that fuel treatment changes both wildfire behavior and downstream damages. Strabo and Reimer (2024) find that local fuel treatments reduce wildfire size and suppression costs, while Strabo et al. (2026) find that treatments reduce wildfire spread and burn severity, limiting structure loss, carbon emissions, and $PM_{2.5}$ exposure.

Fuel treatment can affect wildfire-related health damages through two margins. The *size* margin operates by limiting the area burned after a wildfire ignites. The *severity* margin concerns damage conditional on realized wildfire size, which may be mitigated by fuel treatment through reduced burn severity. In this study, the severity margin is assessed by examining whether the health effect of lightning-induced wildfire size differs with the fuel-treatment area already present in the affected county.[28]

6.2 Wildfire Impacts by Fuel Treatment

This study estimates a county-month IV specification that interacts log natural-caused wildfire size with logged fuel-treatment area at the beginning of the month.[29] The fuel-

[28]The size margin is examined in Strabo et al. (2026), who link treatment-induced reductions in burned area to lower smoke $PM_{2.5}$ exposure and associated health damages. They quantify avoided premature mortality using exposure–mortality estimates from prior research, rather than estimating effects on birth outcomes.

[29]Following Chen and Roth (2024), assigning alternative calibrated values to zero observations after normalizing positive treatment areas by their minimum yields similar birth-weight estimates.

treatment area captures USFS-tracked treatment activity, primarily on National Forest System lands, and does not measure all treatments conducted within a county. Log natural-caused wildfire size and its interaction with logged treatment area are treated as endogenous and instrumented by lightning counts and lightning counts interacted with logged treatment area, respectively.[30]

Table 7 reports a positive interaction for birth weight, consistent with a smaller adverse effect of lightning-induced wildfire size in county-months with larger fuel-treatment area. The low-birth-weight interaction is negative and the gestational-age interaction is positive, but both are imprecisely estimated. The preterm-birth interaction is positive, which is opposite to the direction implied by attenuation, and its magnitude is also uncertain. Appendix Table E.3 shows similar positive birth-weight interactions for mechanical and prescribed-fire treatment areas separately, and Appendix Table E.4 shows a nearly identical estimate after applying annual depreciation to past treatment areas. The birth-weight estimates provide the clearest evidence of conditional attenuation, while the other outcomes do not show a consistent pattern.

Appendix Table E.8 extends the specification to wildfire exposure in the birth month and three and six months before birth, interacting each exposure with treatment area at the beginning of the corresponding month. The birth-weight interactions are positive at all three timings, although their magnitudes remain uncertain, while the other outcomes do not show a consistent attenuation pattern. These estimates are exploratory because exposure timing is indexed to the realized birth month and does not map cleanly into trimesters.

The channel analyses in Appendix E.5 provide complementary evidence consistent with this conditional pattern. Larger fuel-treatment areas are associated with smaller estimated wildfire-related increases in measured smoke exposure (Appendix Table E.5) and maternal tobacco use (Appendix Table E.7), and with a less negative sentiment response (Appendix Table E.6).

6.3 Interpretation and Policy Implications

The positive birth-weight interaction is consistent with a severity margin. Conditional on lightning-induced variation in realized wildfire size, birth-weight damage is smaller where preexisting fuel-treatment area is larger. Because treatment placement is nonrandom, the interaction identifies conditional heterogeneity across preexisting treatment areas rather than the causal effect of expanding treatment.

Assuming that this relationship remains valid when treatment expands, Appendix E.6 translates the implied birth-weight improvement into monetary benefits. Medical-cost

[30] Appendix E.2 describes the construction of treatment area, estimating equation, and IV implementation.

Table 7: Fuel-Treatment Area and Wildfire Effects on Birth Outcomes (IV, county-month level), 2003–2020

	Birth Weight (grams) (1)	LBW per 1,000 Births (2)	LMP Gestational Age (weeks) (3)	Preterm Births per 1,000 Births (4)	SW F (5)
Ln Natural-Caused Fire Size	-0.838***	0.228**	-0.003*	0.418**	235.5
	(0.256)	(0.097)	(0.002)	(0.179)	
Ln Natural-Caused Fire Size × Ln FT Area	0.052**	-0.011	0.0001	0.018	234.3
	(0.021)	(0.007)	(0.0001)	(0.015)	
County FE	X	X	X	X	
Year-Month FE	X	X	X	X	
Weather Controls	X	X	X	X	
FT Area	X	X	X	X	
Mean (Dep. Var.)	3278	79	38.6	117	
SD (Dep. Var.)	76	26	0.3	36	
Counties	576	576	576	576	
Obs.	116,856	116,856	116,856	116,856	
R^2	0.763	0.473	0.585	0.500	

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. Ln FT Area denotes the log of the sum of mechanical and prescribed-fire treatment areas completed before the first day of the month within the county, as defined in Appendix E.2. The implementation adds 10^{-6} km^2 before taking its logarithm. The coefficient on log natural-caused wildfire size is evaluated at a logged treatment area of zero, corresponding to approximately 1 km^2 of treatment area, rather than no treatment. The endogenous variables are log natural-caused wildfire size and its interaction with the logged treatment area. The instruments are lightning counts and lightning counts interacted with the logged treatment area. Each specification includes county fixed effects, year-month fixed effects, the main effect of logged treatment area, log other wildfire size, fire weather controls, pressure, and wind speed, and is weighted by the total number of births. The main effect of logged treatment area is omitted from the table. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row, using the common outcome-estimation sample. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

savings follow the per-gram valuation used by East et al. (2023). The earnings calculation combines the birth-weight elasticity of permanent labor income from Bharadwaj et al. (2018) with the US lifetime earnings estimates of Grosse et al. (2019).

Appendix Figure E.3 shows how these benefits vary with annual wildfire area and the percentage increase in annual treatment area in the 108 counties represented in both the county treatment-area panel and the birth-outcome estimation sample. The left panel reports medical-cost savings, and the right panel adds future labor earnings discounted to birth. At the observed mean annual wildfire area, a 10% expansion of treatment flows implies medical-cost savings of $71.8 thousand per annual birth cohort. Including future earnings raises the benefit to $265.2 thousand. Both benefit measures are expressed in 2020 dollars.

These values sum benefits over births with reported birth weight in the available estimation-sample county-months and divide the total by the 18 years from 2003 to 2020. They capture birth-weight improvements through the severity channel and exclude benefits from reductions in burned area, property damage, and other health losses. Their interpretation depends on the applicability of the external valuations and the assumptions underlying the treatment-expansion scenario.

7 Conclusion

This study uses lightning strikes as an instrument for natural-caused wildfire size to estimate the effects of local wildfire events on birth outcomes in a US county-month panel covering 2003–2020. By exploiting lightning-induced variation in wildfire size, the analysis addresses confounding associated with human ignition behavior and local socioeconomic conditions. In the birth month, a one standard deviation increase in natural-caused wildfire size from its sample mean reduces birth weight by 4.1 grams and increases the incidence of low birth weight by 1.1 per 1,000 births. Estimates for gestational age and preterm birth also point to adverse effects, although their magnitudes are less certain in the full sample. The causal interpretation rests on the assumption that, conditional on the controls and fixed effects, lightning affects birth outcomes through natural-caused wildfire size.

The analysis then examines the pathways linking local wildfire events to birth outcomes. Adding HMS smoke coverage and wildfire-specific $PM_{2.5}$ controls leaves the estimated wildfire-size effects largely unchanged. Wildfires also lower Twitter sentiment, increase maternal tobacco use, and cause greater reported building involvement and property losses. Together with the demographic heterogeneity results, these patterns provide suggestive evidence of a local-disruption channel that includes stress-related responses. They do not identify the contribution of stress to the birth-outcome effects or rule out an air-pollution pathway. The findings therefore complement the evidence on wildfire smoke by documenting health risks associated with local wildfire events.

Finally, the adverse birth-weight effect of lightning-induced wildfire size is smaller in counties with larger preexisting fuel-treatment areas. Complementary patterns for measured smoke exposure, sentiment, and maternal tobacco use are consistent with this attenuation. Because treatment placement is nonrandom, these results describe conditional heterogeneity rather than the causal effect of expanding fuel treatment. The illustrative valuation translates the implied birth-weight improvements into medical-cost savings and future earnings under assumptions about treatment expansion and the applicability of external valuations. These findings highlight birth outcomes as a relevant dimension of wildfire damages and a potential benefit to consider in evaluating wildfire-management policies.

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A Supplementary Summary Statistics

Table A.1 reports summary statistics for maternal tobacco use and NFIRS outcomes.

Table A.1: Summary Statistics for Maternal Tobacco Use and NFIRS Outcomes

	Obs.	Mean	SD	Min	P10	P90	Max
Panel A. Maternal tobacco use							
Maternal Tobacco Use per 1,000 Births with Reported Tobacco Status	72,149	93.38	67.68	0	17.54	188.03	460.32
Panel B. NFIRS-reported outcomes							
Buildings Involved per 100,000 Population	116,856	4.28	6.33	0	0.44	8.74	710.96
Property Loss per Resident (Nominal USD)	116,856	1.53	26.20	0	0.00	2.83	6,090.84
Content Loss per Resident (Nominal USD)	116,856	0.60	15.09	0	0.00	1.03	4,344.85
Civilian Casualties per 100,000 Population	116,856	0.32	0.68	0	0.00	0.93	119.64
Respiratory Casualties per 100,000 Population	116,856	0.09	0.30	0	0.00	0.29	18.79
Burn Casualties per 100,000 Population	116,856	0.10	0.26	0	0.00	0.37	5.49

Notes: This table reports unweighted summary statistics, including the number of nonmissing county-month observations (Obs.), mean, standard deviation (SD), minimum (Min), tenth percentile (P10), ninetieth percentile (P90), and maximum (Max). Panel A reports maternal tobacco use per 1,000 births with a reported tobacco status for county-months meeting the 95% reporting-coverage restriction over 2007–2020. Panel B reports all NFIRS outcomes for 2003–2020. Buildings involved and civilian casualties are measured per 100,000 residents. Property and content losses are nominal dollars per resident. The corresponding regressions use $\ln(1+x)$. The unconditional NFIRS construction assigns zero when a county-month has no matched NFIRS record. CDC-suppressed tobacco-use counts of nine or fewer are imputed as 5.

B Literature Review

Table B.1: Summary of Wildfire and Air Pollution Studies

Category	Study	Wildfire Measure	Outcome
1. Smoke Measures	Fadadu et al. (2021)	HMS smoke plume density	AD/itch clinic visits
	Heft-Neal et al. (2022)	HMS-derived monthly smoke exposure	Preterm birth risk
	Borgschulte et al. (2024)	HMS smoke plume coverage	Earnings and employment
	Wen and Burke (2022)	HMS-derived smoke $PM_{2.5}$	Standardized math and English test scores
	Molitor et al. (2023)	HMS smoke plume density	Suicide
	Cabral and Dillender (2024)	HMS smoke plume coverage	Injury claims
	Miller et al. (2024)	HMS smoke plume proximity and density	Mortality and emergency room visits
	Lee and Beatty (2025)	HMS smoke plume coverage / density	Agricultural worker movement
2. Wind and Dispersion	Jayachandran (2009)	Wind-direction smoke exposure	Infant mortality
	Moeltner et al. (2013)	Wind-direction weighted acres burned	Cardiorespiratory admissions
	Sheldon and Sankaran (2017)	Fire radiative power × wind speed	Clinic visits
	He et al. (2020)	Upwind vs. downwind fires	Mortality
	Graff Zivin et al. (2020)	Upwind vs. downwind fires	Test scores
	Arrizaga et al. (2026)	HYSPLIT model	ED visits, dropout rates, and birth outcomes
	Zhang (2026)	Upwind fire	Child development
3. Proximity	Currie and Saberian (2025)	Fires within 100 km	Mental health
	O'Donnell and Behie (2013)	Residence in fire areas	Birth weight
	Korsiak et al. (2022)	Burned area $< 20/50$ km	Lung and brain cancer

Notes: Studies are grouped by exposure measure into smoke, wind and dispersion, and proximity. The table reports selected outcomes. HMS denotes Hazard Mapping System, AD denotes atopic dermatitis, and ED denotes emergency department. HYSPLIT denotes Hybrid Single-Particle Lagrangian Integrated Trajectory model.

Table B.2: Summary of In-Utero Stress Literature

Category	Study	Stress Measure	Outcome	Timing
1. Natural Disasters	Kim et al. (2017)	Northridge earthquake proximity	Birth weight, LBW	All trimesters (strongest in 1st or 3rd)
	de Oliveira et al. (2023)	Hurricane Catarina	Birth weight, LBW, fetal mortality	During pregnancy (pooled)
2. Mass Tragedies	Brown (2020)	9/11 attacks (gestational timing)	Birth weight, fetal growth, preterm birth	Early pregnancy (fetal growth) Mid-pregnancy (preterm birth)
	Quintana-Domeque and Ródenas-Serrano (2017)	ETA terrorism (bomb casualties)	Birth weight, LBW	1st trimester (strongest effects)
3. Economic Shocks	Carlson (2015)	Mass layoff announcements (WARN)	Birth weight, LBW	Late pregnancy (strongest near birth)
4. Violence & Crime	Currie et al. (2022)	Prenatal assault (crime records × address)	Birth weight, preterm birth	3rd trimester (strongest effects)
5. Bereavement	Black et al. (2016)	Parental death during pregnancy (sibling FE)	Birth weight	Across pregnancy (no clear timing pattern)
	Persson and Rossin-Slater (2018)	Family bereavement during pregnancy	ADHD medication	Across pregnancy (no clear timing pattern)
6. Stress Biomarkers	Aizer et al. (2016)	Maternal cortisol levels (mother FE)	Cognition, chronic illness, education	3rd trimester (measured at 31–36 weeks)

Notes: This table summarizes selected studies linking prenatal stressors or maternal cortisol to offspring outcomes. Studies are grouped by stressor type or the use of a stress biomarker. The Outcome column lists selected outcomes examined in each study, including outcomes measured after birth. The Timing column distinguishes reported patterns of stronger estimated effects from exposure or measurement windows. Exposure is pooled across pregnancy in de Oliveira et al. (2023). For Brown (2020), fetal growth refers to birth weight adjusted for gestational age. For Aizer et al. (2016), the timing entry describes cortisol measurement at 31–36 weeks, rather than the trimester of greatest vulnerability. LBW denotes low birth weight, FE denotes fixed effects, and ADHD denotes attention deficit hyperactivity disorder. WARN refers to the Worker Adjustment and Retraining Notification Act. ETA denotes Euskadi ta Askatasuna (Basque Homeland and Freedom).

C Robustness Checks

C.1 Obstetric-Estimate Gestational Age

Table C.1 compares the main 2003–2020 LMP specification with LMP and OE estimates over the common 2007–2020 period. The OE estimates retain the adverse signs for gestational age and preterm birth and have smaller standard errors than the LMP estimates over either period. Restricting LMP to 2007–2020 does not produce comparable precision gains, so the period restriction alone does not explain the smaller standard errors under OE. This pattern is consistent with LMP measurement error contributing to uncertainty in the main estimates.

Table C.1: Gestational-Age Definition Robustness (IV, county-month level), 2003–2020 and 2007–2020

	LMP 2003–2020 (1)	LMP 2007–2020 (2)	OE 2007–2020 (3)
Panel A: Average Gestational Age (weeks)			
Ln Natural-Caused Fire Size	-0.0035*	-0.0018	-0.0028**
	(0.0018)	(0.0018)	(0.0013)
Mean (Dep. Var.)	38.620	38.626	38.480
SD (Dep. Var.)	0.305	0.304	0.264
Panel B: Preterm Births per 1,000 Births			
Ln Natural-Caused Fire Size	0.272	0.202	0.348**
	(0.197)	(0.210)	(0.138)
Mean (Dep. Var.)	117.183	115.427	96.930
SD (Dep. Var.)	35.721	36.040	28.893
County FE	X	X	X
Year-Month FE	X	X	X
Weather Controls	X	X	X
F^{Eff}	49.97	40.42	40.42
Counties	576	576	576
Obs.	116,856	91,896	91,896

Notes: Panel A reports county average gestational age in weeks, and Panel B reports births below 37 completed weeks per 1,000 births with gestational age reported under the corresponding definition. Column 1 uses last-menstrual-period (LMP) gestational age over 2003–2020. Column 2 restricts the LMP specification to 2007–2020, and Column 3 uses the obstetric estimate (OE) over the same 2007–2020 period. The observational unit is the county-month. All columns report IV estimates that instrument log natural-caused wildfire size with lightning counts. All estimations include log other wildfire size, county fixed effects, year-month fixed effects, fire weather controls, pressure, and wind speed and are weighted by the total number of births, including births with missing outcome-reporting status. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic and is reported once per column because the two outcomes use the same estimation sample within each specification. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics for the outcome definition and estimation sample in each column. Significance stars follow the conventional notation.

C.2 First-Stage Heterogeneity Robustness

With a heterogeneous first stage, additive two-stage least squares (2SLS) need not identify a non-negatively weighted average of conditional effects (Słoczyński, 2020). The stronger lightning–fire relationship under hot and dry conditions motivates this sensitivity check (Table 3, Column 3). Following Angrist and Imbens (1995), the specification instruments log natural-caused wildfire size with lightning counts and their interactions with log precipitation, log specific humidity, and log air temperature. Table C.2 compares this four-instrument specification with the single-instrument baseline. The estimates retain the baseline signs across all four outcomes and are smaller in magnitude for birth weight, low birth weight, and gestational age, while the preterm-birth estimate changes less.

Table C.2: Wildfires and Birth Outcomes with Weather-Interacted Instruments (IV, county-month level), 2003–2020

	Birth Weight (grams)		Low Birth Weight per 1,000 Births		LMP Gestational Age (weeks)		Preterm Births per 1,000 Births	
	Baseline (1)	Interacted (2)	Baseline (3)	Interacted (4)	Baseline (5)	Interacted (6)	Baseline (7)	Interacted (8)
Ln Natural-Caused Fire Size	-1.219***	-0.575**	0.311**	0.105	-0.004*	-0.002	0.272	0.325***
	(0.334)	(0.270)	(0.122)	(0.089)	(0.002)	(0.001)	(0.197)	(0.118)
County FE	X	X	X	X	X	X	X	X
Year-Month FE	X	X	X	X	X	X	X	X
Fire Weather Controls	X	X	X	X	X	X	X	X
No. of Instruments	1	4	1	4	1	4	1	4
F^{Eff}	49.97	27.40	49.97	27.40	49.97	27.40	49.97	27.40
Mean (Dep. Var.)	3278	3278	79	79	38.6	38.6	117	117
SD (Dep. Var.)	76	76	26	26	0.3	0.3	36	36
Counties	576	576	576	576	576	576	576	576
Obs.	116,856	116,856	116,856	116,856	116,856	116,856	116,856	116,856
R^2	0.764	0.766	0.474	0.475	0.586	0.587	0.500	0.500

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. The "Baseline" columns instrument log natural-caused wildfire size with the lightning count, as in Table 5. The "Interacted" columns additionally interact the lightning instrument with the three base fire weather variables (log precipitation, log specific humidity, and log air temperature), yielding four instruments for the single endogenous fire-size variable, so that the first stage is allowed to vary with weather (Angrist and Imbens, 1995). All estimations include log other wildfire size, county fixed effects, year-month fixed effects, fire weather controls, pressure, and wind speed and are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. F^{Eff} is the outcome-specific effective first-stage F-statistic. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

C.3 Exposure Timing and Expanding Exposure Windows

Table C.3 reports the distributed-lag IV specification, and Table C.4 reports the specification based on average exposure over expanding windows, as discussed in Section 5. Adding a nine-month lag to the distributed-lag specification materially increases the standard errors. Each lag requires a separate lightning instrument, and the lagged lightning counts share strong seasonal patterns, reducing their conditional identifying variation as lags are added. Both sets of estimates remain exploratory because exposure lags are indexed to realized birth month and instrument strength declines as the exposure window extends.

Table C.3: Exploratory Timing Specification for Wildfires and Birth Outcomes (IV, county-month level), 2003–2020

	Birth Weight		LMP Gestational Age		SW F-statistic
	Birth Weight (grams) (1)	LBW per 1,000 Births (2)	LMP Gestational Age (weeks) (3)	Preterm Births per 1,000 Births (4)	SW F (5)
Ln Natural-Caused Fire Size	-1.426**	0.239	-0.005	-0.075	289.8
	(0.702)	(0.222)	(0.004)	(0.461)	
Ln Natural-Caused Fire Size, 3 Months Before Birth	-1.317*	0.087	-0.008*	-0.207	125.7
	(0.674)	(0.216)	(0.004)	(0.454)	
Ln Natural-Caused Fire Size, 6 Months Before Birth	-1.495*	-0.163	-0.008	-0.461	122.5
	(0.896)	(0.275)	(0.006)	(0.663)	
County FE	X	X	X	X	
Year-Month FE	X	X	X	X	
Fire Weather Controls	X	X	X	X	
Mean (Dep. Var.)	3278	79	38.6	117	
SD (Dep. Var.)	76	26	0.3	36	
Counties	576	576	576	576	
Obs.	116,856	116,856	116,856	116,856	
R^2	0.755	0.474	0.566	0.499	

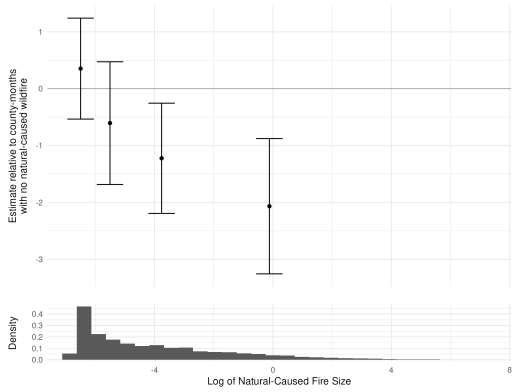
Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. All columns report IV estimates of log natural-caused wildfire size in the birth month and three and six months before birth, instrumented by lightning counts in the corresponding months. All estimations include county fixed effects, year-month fixed effects, and contemporaneous and corresponding lagged log other wildfire size, fire weather controls, pressure, and wind speed. Estimations are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) conditional first-stage F-statistic (Sanderson and Windmeijer, 2016) on the corresponding endogenous-variable coefficient row, using the common outcome-estimation sample. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

Table C.4: Average Wildfire Exposure over Expanding Windows and Birth Outcomes (IV, county-month level), 2003–2020

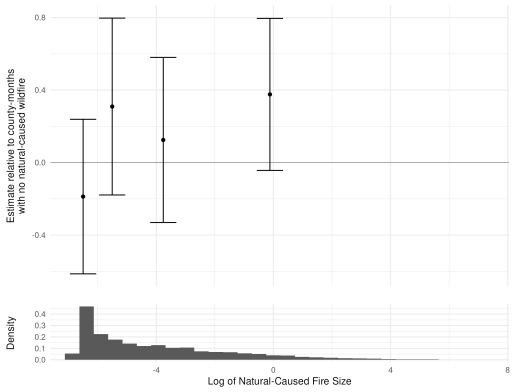
	0–2 Months			0–5 Months			0–9 Months			
	0	1	2	3	4	5	6	7	8	9
Panel A: Birth Weight (grams)										
Ln (Average Natural-Caused Fire Size)	-1.219***	-1.262***	-1.532***	-1.924***	-2.219***	-2.546**	-2.794**	-2.799	-2.518	-3.362
	(0.334)	(0.388)	(0.495)	(0.650)	(0.797)	(1.013)	(1.335)	(1.845)	(2.795)	(5.132)
R^2	0.764	0.764	0.762	0.758	0.755	0.752	0.750	0.751	0.755	0.746
Panel B: Low Birth Weight per 1,000 Births										
Ln (Average Natural-Caused Fire Size)	0.311**	0.232**	0.333**	0.401**	0.421*	0.579**	0.466	0.091	-0.800	-2.285
	(0.122)	(0.118)	(0.150)	(0.189)	(0.218)	(0.283)	(0.353)	(0.466)	(0.820)	(2.134)
R^2	0.474	0.474	0.473	0.471	0.471	0.468	0.470	0.475	0.462	0.373
Panel C: LMP Gestational Age (weeks)										
Ln (Average Natural-Caused Fire Size)	-0.004*	-0.004*	-0.006**	-0.009***	-0.011**	-0.013**	-0.014*	-0.017	-0.019	-0.029
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.006)	(0.008)	(0.012)	(0.021)	(0.039)
R^2	0.586	0.585	0.581	0.574	0.568	0.562	0.555	0.545	0.534	0.478
Panel D: Preterm Births per 1,000 Births										
Ln (Average Natural-Caused Fire Size)	0.272	0.257	0.381	0.425	0.320	0.125	-0.242	-0.887	-2.395	-4.717
	(0.197)	(0.208)	(0.259)	(0.338)	(0.420)	(0.569)	(0.792)	(1.235)	(2.316)	(5.240)
R^2	0.500	0.501	0.499	0.499	0.500	0.501	0.500	0.492	0.439	0.273
Panel E: First stage										
Average of Lightning Count	0.0001***	0.0002***	0.0002***	0.0002***	0.0002***	0.0002***	0.0001***	0.0001***	0.0001**	0.0001
	(0.00002)	(0.00002)	(0.00003)	(0.00003)	(0.00003)	(0.00003)	(0.00004)	(0.00004)	(0.00004)	(0.00004)
R^2	0.457	0.532	0.570	0.596	0.615	0.628	0.643	0.660	0.679	0.695
F^{Eff}	49.97	49.05	43.54	38.00	32.10	24.09	15.96	9.50	4.69	1.74
Obs.	116,856	116,856	116,856	116,856	116,856	116,856	116,856	116,856	116,856	116,856

Notes: Each column k presents a 2SLS estimate where the endogenous variable is the log of average natural-caused wildfire size from the birth month ($t = 0$) through k months prior to birth, and the instrument is the average lightning count over the same window. The observational unit is the county-month. Panels A–D report the second-stage estimates for county average birth weight, low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age, and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age, respectively. Panel E reports the first stage for the common outcome-estimation sample. Fire weather controls are log precipitation, log specific humidity, log air temperature, and their interactions. Other controls are log other wildfire size, pressure, and wind speed. All controls are averaged over the same window. All specifications include county fixed effects and year-month fixed effects, and are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Significance stars follow the conventional notation.

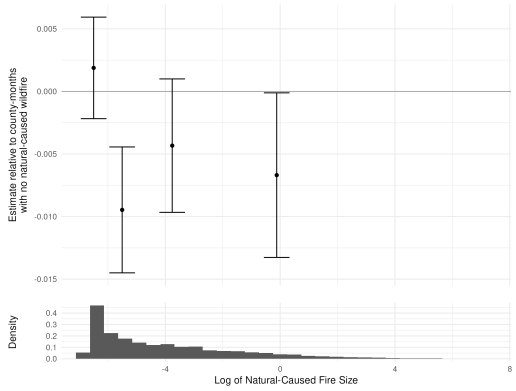
C.4 Dose-Response Functions



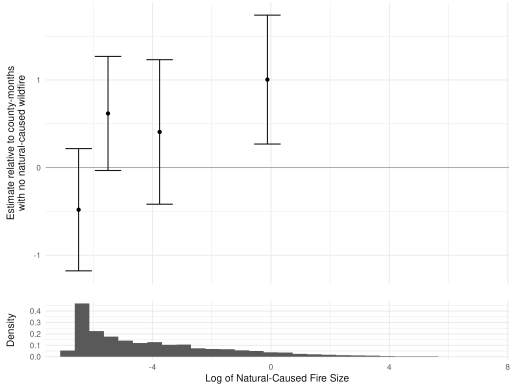
Ave. Birth Weight (grams)



Low-Birth-Weight Rate (per 1,000 births)



Ave. LMP Gestational Age (weeks)



LMP Preterm-Birth Rate (per 1,000 births)

Figure C.1: Binned Dose-Response Diagnostics for Wildfire Exposure and Birth Outcomes

Note: The estimation sample covers 2003–2020. The upper panel of each subfigure reports coefficients on indicators for quartiles of positive natural-caused wildfire size, relative to county-months with no natural-caused wildfire. Points are located at the birth-weighted mean log wildfire size within each quartile, and vertical bars show 95% confidence intervals based on standard errors clustered by county. The lower panels show the unweighted distribution of log wildfire size among positive-fire county-months. All estimations include county and year-month fixed effects, log other wildfire size (combining identified human-caused wildfires and wildfires with undetermined or unclassified causes), fire weather controls, pressure, and wind speed and are weighted by the total number of births. Gestational age uses the LMP definition. The preterm-birth rate is measured per 1,000 births with reported LMP gestational age.

C.5 Additional Controls and Fixed Effects

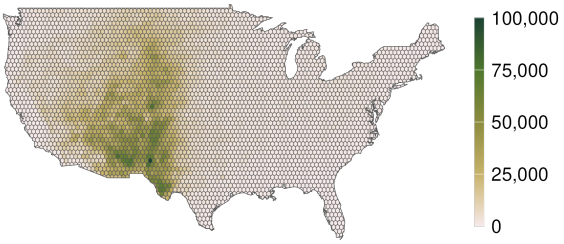
Table C.5: Wildfires and Birth Outcomes with Additional Controls and Fixed Effects (IV, county-month level), 2003–2020

	Baseline model	Birth Controls	Maternal Controls	Storm Controls	Season FE × County FE	Lagged Weather	County FE × Year FE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Birth Weight (grams)							
Ln Natural-Caused Fire Size	-1.219***	-1.105***	-1.117***	-1.445***	-1.750***	-1.444***	-0.666**
	(0.334)	(0.330)	(0.312)	(0.333)	(0.654)	(0.392)	(0.268)
R^2	0.764	0.787	0.773	0.763	0.766	0.763	0.807
Panel B: Low Birth Weight per 1,000 Births							
Ln Natural-Caused Fire Size	0.311**	0.268**	0.348***	0.394***	0.313	0.403***	0.283**
	(0.122)	(0.115)	(0.121)	(0.133)	(0.266)	(0.145)	(0.117)
R^2	0.474	0.534	0.479	0.472	0.482	0.472	0.531
Panel C: LMP Gestational Age (weeks)							
Ln Natural-Caused Fire Size	-0.004*	-0.003*	-0.003*	-0.004**	-0.006	-0.004*	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.004)	(0.002)	(0.001)
R^2	0.586	0.607	0.592	0.585	0.591	0.585	0.707
Panel D: Preterm Births per 1,000 Births							
Ln Natural-Caused Fire Size	0.272	0.235	0.330*	0.380*	0.386	0.295	0.268*
	(0.197)	(0.199)	(0.196)	(0.213)	(0.467)	(0.247)	(0.145)
R^2	0.500	0.531	0.505	0.500	0.509	0.501	0.619
Panel E: First Stage							
Std. Lightning Count	0.591***	0.590***	0.597***	0.569***	0.310***	0.491***	0.649***
	(0.084)	(0.083)	(0.085)	(0.077)	(0.058)	(0.075)	(0.091)
R^2	0.457	0.457	0.458	0.457	0.543	0.471	0.529
F^{Eff}	49.97	50.24	49.72	55.10	29.41	42.66	56.13
Obs.	116,856	116,856	116,856	116,856	116,856	116,856	116,856

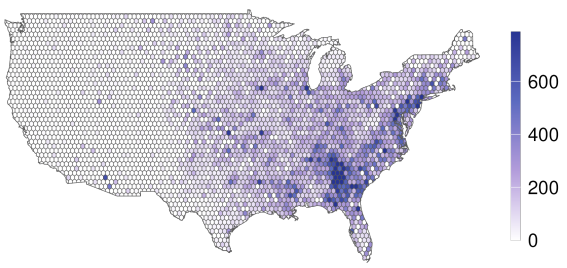
Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. Panels A–D report IV estimates of log natural-caused wildfire size instrumented by lightning counts. All estimations include log other wildfire size, fire weather controls, pressure, wind speed, county fixed effects, and year-month fixed effects. Column 1 reports this baseline specification. Columns 2 and 3 add infant and birth characteristics (medical attendant at birth, infant sex, birthplace, and multiple births) and maternal characteristics (average age, live-birth order, Hispanic origin, race, and marital status), respectively. Column 4 adds storm events and property damage. Column 5 adds interactions of county fixed effects with indicators for January–March, April–June, July–September, and October–December. Columns 6 and 7 add one-month-lagged fire weather controls and county-by-year fixed effects, respectively. Estimations are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. Panel E reports the first-stage coefficient on lightning count standardized over the complete 2003–2020 county-month panel. The first-stage models use the same estimation sample and weights as Panels A–D. F^{Eff} is the effective first-stage F-statistic computed on that common sample. Significance stars follow the conventional notation.

C.6 Dry Lightning and Storm Events

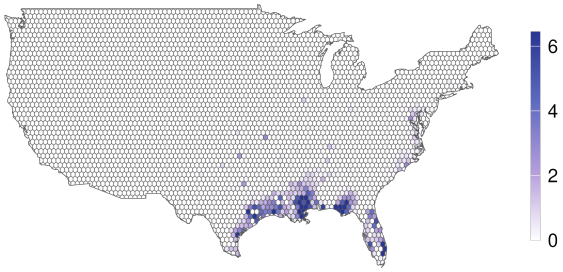
Storm data come from the NOAA Storm Events Database for 2003–2020. County-reported events are matched by Federal Information Processing Standards (FIPS) code, while events reported by National Weather Service forecast zone are assigned by intersecting forecast-zone polygons with county exposure areas. The number of assigned events and their full reported property damage are summed by county and month for the storm controls in Column 4 of Table C.5. Figure C.2 shows that lightning strikes under low-precipitation and low-humidity conditions are concentrated in the southwestern and interior western US, whereas storm events and property damage are concentrated in the southeastern US and along the Gulf Coast.



Lightning Strikes under Low-Precipitation and Low-Humidity Conditions (count)



Storm Events (count)



Storm Property Damage (billion USD)

Figure C.2: Geographic Distribution of Dry Lightning, Storm Events, and Storm Property Damage

Note: All panels summarize observations over 2003–2020. The contiguous-US mainland is divided into a regular hexagonal grid with 0.5 degrees between opposite edges. Detached islands are excluded, while the grid fully covers the mainland polygon. Hexagonal cells along the coast are clipped to the contiguous-US mainland coastline for display. Every retained lightning location is assigned to one hexagon. The top panel shows the total number of dry lightning strikes, defined as lightning strikes occurring when daily precipitation and specific humidity are below their respective sample medians among hexagon-days with recorded lightning during 2003–2020. The bottom panels show the total number of storm events and total property damage, respectively, assigned to counties and aggregated across thunderstorm wind, tornado, lightning, hurricane (typhoon), and tropical storm events from the NOAA Storm Events Database. County-reported events are assigned by FIPS code, while forecast-zone records are assigned by intersection with county exposure areas.

C.7 Robustness to Power-Outage Controls

Lightning-related outages pose a threat to the exclusion restriction, and outages have been linked to fertility and birth weight in developing-country settings (Burlando, 2014a,b). The analysis uses county-level EAGLE-I data from Oak Ridge National Laboratory, recorded at fifteen-minute intervals, for November 2014–December 2020. Table C.6 adds an indicator for any outage, the average share of customers without power, and the log of one plus total customer-outage hours to the baseline IV specification, one at a time and jointly. Intervals absent from the positive-outage records are assigned zero when constructing the monthly measures. Coverage is incomplete, especially in the early years, so an absent record may represent either zero outages or a collection gap.

The estimates retain their baseline signs, while the birth-weight and preterm-birth estimates attenuate when all three outage controls are included. Standard errors are larger in this smaller subsample than in the full sample. Outages may also be a downstream consequence of wildfire, so conditioning on them does not have a simple causal interpretation. Given these limitations, the estimates provide only a limited check on a lightning–outage–health pathway.

Table C.6: Wildfires and Birth Outcomes with Power-Outage Controls (IV, county-month level), November 2014–December 2020

	Baseline	+ Any Outage	+ Outage Share	+ Outage Hours	+ All
	(1)	(2)	(3)	(4)	(5)
Panel A: Birth Weight (grams)					
Ln Natural-Caused Fire Size	-0.288	-0.282	-0.294	-0.246	-0.207
	(0.505)	(0.504)	(0.507)	(0.507)	(0.518)
R^2	0.757	0.757	0.757	0.757	0.757
Panel B: Low Birth Weight per 1,000 Births					
Ln Natural-Caused Fire Size	0.365*	0.365*	0.375*	0.361*	0.372*
	(0.200)	(0.201)	(0.202)	(0.201)	(0.203)
R^2	0.466	0.466	0.466	0.466	0.466
Panel C: LMP Gestational Age (weeks)					
Ln Natural-Caused Fire Size	-0.001	-0.001	-0.001	-0.001	-0.001
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
R^2	0.605	0.605	0.605	0.605	0.605
Panel D: Preterm Births per 1,000 Births					
Ln Natural-Caused Fire Size	0.397	0.380	0.395	0.325	0.274
	(0.310)	(0.314)	(0.311)	(0.326)	(0.333)
Obs.	42,476	42,476	42,476	42,476	42,476
F^{Eff}	33.51	33.55	33.76	33.37	33.46
R^2	0.540	0.541	0.540	0.541	0.542

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. Column 1 reports the baseline specification. Columns 2–4 add an indicator for any outage, the average share of customers without power, and the log of one plus total customer-outage hours, respectively. Column 5 includes all three outage controls. To construct the average monthly share, the analysis divides each fifteen-minute outage count by the supplied 2022 modeled county customer count, caps the resulting share at one, assigns zero to intervals absent from the positive-outage records, and averages over all intervals in the month. Because an absent source record may indicate either zero outages or a collection gap, outage coverage is incomplete, especially in the early years. The observational unit is the county-month, restricted to November 2014–December 2020. All estimations include log other wildfire size, log precipitation, log specific humidity, log air temperature, their pairwise and three-way interactions, pressure, wind speed, county fixed effects, and year-month fixed effects. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Log natural-caused wildfire size is instrumented with lightning strikes. Estimations are weighted by the total number of births. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Significance stars follow the conventional notation.

C.8 Log with Zeros

Table C.7: Wildfires and Birth Outcomes with Alternative Log Transformations (IV, county-month level), 2003–2020

	Baseline	x = 0	x = -0.1	x = -1	x = -3	x = -5
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Birth Weight (grams)						
Ln Natural-Caused Fire Size	-1.219***	-1.359***	-1.348***	-1.256***	-1.091***	-0.964***
	(0.334)	(0.373)	(0.370)	(0.344)	(0.298)	(0.263)
R^2	0.764	0.764	0.764	0.764	0.764	0.764
Panel B: Low Birth Weight per 1,000 Births						
Ln Natural-Caused Fire Size	0.311**	0.347**	0.344**	0.320**	0.278**	0.246**
	(0.122)	(0.136)	(0.135)	(0.126)	(0.109)	(0.096)
R^2	0.474	0.474	0.474	0.474	0.473	0.473
Panel C: LMP Gestational Age (weeks)						
Ln Natural-Caused Fire Size	-0.004*	-0.004*	-0.004*	-0.004*	-0.003*	-0.003*
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
R^2	0.586	0.586	0.586	0.586	0.585	0.585
Panel D: Preterm Births per 1,000 Births						
Ln Natural-Caused Fire Size	0.272	0.303	0.301	0.280	0.243	0.215
	(0.197)	(0.220)	(0.218)	(0.203)	(0.176)	(0.156)
R^2	0.500	0.501	0.501	0.501	0.500	0.500
F^{Eff}	49.97	48.71	48.82	49.64	51.08	52.12
Obs.	116,856	116,856	116,856	116,856	116,856	116,856

Notes: For the alternative columns, positive natural-caused wildfire size is divided by its minimum positive value before taking logs, and zero-fire observations are assigned the value shown in the column heading ($x = 0, -0.1, -1, -3, -5$). The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. All columns report IV estimates that instrument the corresponding transformed natural-caused wildfire-size measure with lightning counts. All estimations include county fixed effects, year-month fixed effects, log other wildfire size, fire weather controls, pressure, and wind speed and are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Significance stars follow the conventional notation.

C.9 Spatial Clustering Robustness

Table C.8: Robustness Checks for Different Levels of Clustering (OLS, county-month level), 2003–2020

Clustering Variables	No. of Clusters	Outcome			
		Birth Weight (grams)	LBW per 1,000 Births	LMP Gestational Age (weeks)	Preterm Births per 1,000 Births
		-0.118	0.022	-0.0005	0.058
County	576	(0.037) ***	(0.013)	(0.0002) **	(0.025) **
County + Year-Month	576 + 216	(0.042) ***	(0.014)	(0.0002) **	(0.029) **
County + Year	576 + 18	(0.040) ***	(0.012) *	(0.0003)	(0.032) *
Conley (50 km)	-	(0.038) ***	(0.013) *	(0.0002) **	(0.027) **
Conley (100 km)	-	(0.041) ***	(0.014)	(0.0002) ***	(0.027) **
Conley (150 km)	-	(0.042) ***	(0.014)	(0.0002) ***	(0.031) *

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. All estimations include county fixed effects, year-month fixed effects, log other wildfire size, fire weather controls, pressure, and wind speed and are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. The first coefficient row reports the OLS estimate. Subsequent rows report standard errors using the clustering or spatial-correlation correction indicated in the first column. Conley rows use 50, 100, or 150 km distance cutoffs. Significance stars follow the conventional notation.

Table C.9: Robustness Checks for Different Levels of Clustering (IV, county-month level), 2003–2020

Clustering Variables	No. of Clusters	Outcome				F^{Eff}
		Birth Weight (grams)	LBW per 1,000 Births	LMP Gestational Age (weeks)	Preterm Births per 1,000 Births	
		-1.219	0.311	-0.004	0.272	
County	576	(0.334) ***	(0.122) **	(0.002) *	(0.197)	49.97
County + Year-Month	576 + 216	(0.430) ***	(0.129) **	(0.002) **	(0.221)	34.63
County + Year	576 + 18	(0.378) ***	(0.103) ***	(0.002) **	(0.184)	34.92
Conley (50 km)	-	(0.339) ***	(0.123) **	(0.002) *	(0.214)	41.20
Conley (100 km)	-	(0.342) ***	(0.128) **	(0.002)	(0.214)	28.66
Conley (150 km)	-	(0.349) ***	(0.130) **	(0.002)	(0.201)	23.80

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. The IV estimate instruments log natural-caused wildfire size with lightning counts. All estimations include county fixed effects, year-month fixed effects, log other wildfire size, fire weather controls, pressure, and wind speed and are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. The first coefficient row reports the IV estimate. Subsequent rows report standard errors using the clustering or spatial-correlation correction indicated in the first column. Conley rows use 50, 100, or 150 km distance cutoffs. F^{Eff} is the effective first-stage F-statistic. Significance stars follow the conventional notation.

C.10 Post-Birth Placebo

Table C.10: Post-Birth Placebo Test (IV, county-month level), January 2003–September 2020

	Birth Weight		LMP Gestational Age		SW F-statistic
	Birth Weight (grams)	Low Birth Weight per 1,000 Births	LMP Gestational Age (weeks)	Preterm Births per 1,000 Births	SW F
	(1)	(2)	(3)	(4)	(5)
Ln Natural-Caused Fire (t)	-1.311**	0.193	-0.003	-0.021	201.8
	(0.624)	(0.235)	(0.004)	(0.392)	
Ln Natural-Caused Fire ($t+1$)	-0.717	0.097	-0.002	0.316	216.6
	(0.527)	(0.253)	(0.003)	(0.354)	
Ln Natural-Caused Fire ($t+2$)	0.256	0.022	0.001	0.070	212.2
	(0.531)	(0.273)	(0.002)	(0.297)	
Ln Natural-Caused Fire ($t+3$)	-0.818	-0.098	-0.002	-0.120	193.8
	(0.616)	(0.255)	(0.003)	(0.377)	
County FE	X	X	X	X	
Year-Month FE	X	X	X	X	
Weather Controls	X	X	X	X	
Joint placebo Wald p-value	0.061	0.963	0.525	0.705	
Mean (Dep. Var.)	3278	78	38.6	117	
SD (Dep. Var.)	76	26	0.3	36	
Counties	576	576	576	576	
Obs.	115,134	115,134	115,134	115,134	
R^2	0.762	0.475	0.587	0.503	

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. The outcome-month sample runs from January 2003 through September 2020 because constructing the three post-birth horizons requires fire, lightning, and weather observations through December 2020. Log natural-caused wildfire size at months $t+1$, $t+2$, and $t+3$ is added as a placebo endogenous variable alongside the contemporaneous fire term (t). Each endogenous variable is instrumented by the corresponding lightning count. All specifications include log other wildfire size, fire weather controls, pressure, and wind speed at t through $t+3$, county fixed effects, and year-month fixed effects and are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row, using the common outcome-estimation sample. The joint placebo Wald p-value row reports the unadjusted p-value for the null that the three post-birth coefficients are simultaneously zero. The family-wise Bonferroni threshold across the four outcomes is 0.0125. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

D Pathway of Impact: Wildfire Smoke and Wildfire Events

Table D.1: Summary Statistics for Smoke Controls and Daily Channel Outcomes

	Obs.	Mean	SD	Min	P10	P90	Max
Panel A. Smoke controls (county-month)							
Wildfire Smoke $PM_{2.5}$ ($\mu g/m^3$)	98,136	0.39	1.51	0.00	0.00	1.04	96.43
HMS Smoke Area-Days (km^2-days)	98,136	18,082.05	36,413.85	0.00	0.00	58,884.91	1,240,235.82
Panel B. Smoke outcomes (county-day)							
Wildfire Smoke $PM_{2.5}$ ($\mu g/m^3$)	3,155,904	0.39	2.73	0.00	0.00	0.06	394.41
HMS Smoke Area (km^2)	3,155,904	591.80	2,108.26	0.00	0.00	295.94	52,072.01
Panel C. Twitter sentiment (county-day)							
Twitter Sentiment (Standardized)	1,220,070	0.00	1.00	-18.59	-1.18	1.20	15.41

Notes: This table reports unweighted summary statistics, including observations (Obs.), mean, standard deviation (SD), minimum (Min), tenth percentile (P10), ninetieth percentile (P90), and maximum (Max). Panel A uses the common county-month estimation sample for Table D.3 during 2006–2020. Wildfire smoke $PM_{2.5}$ is the monthly average concentration, including days without reported smoke. HMS smoke area-days sum daily smoke-covered area within the county exposure area over the month. Both controls enter the regressions as the log of one plus the reported measure. Panel B uses the county-day estimation sample for Table E.5 from January 1, 2006, through December 31, 2020. Daily wildfire smoke $PM_{2.5}$ and HMS smoke area are reported in levels. These outcomes also enter the regressions in Table E.5 in levels. Unreported wildfire-smoke $PM_{2.5}$ county-days are assigned zero within the coverage period. Panel C uses the county-day estimation sample for Table D.4 from May 22, 2014, through December 31, 2020, and reports daily Twitter sentiment standardized using the unweighted mean and standard deviation of its estimation sample. County-days without a geolocated post remain missing. All panels cover the 576 counties in the main birth-outcome sample, with observations selected for the corresponding regressions.

D.1 Smoke Exposure Outcomes

Table D.2: Wildfires and Smoke Exposure (IV, county-month level), 2006–2020

	HMS Smoke Area-Days	Wildfire Smoke $PM_{2.5}$
	(1)	(2)
Ln Natural-Caused Fire Size	2985.301***	0.118***
	(672.502)	(0.020)
County FE	X	X
Year-Month FE	X	X
Weather Controls	X	X
F^{Eff}	159.36	159.36
Mean (Dep. Var.)	18013.803	0.393
SD (Dep. Var.)	36188.307	1.487
Counties	576	576
Obs.	103,680	103,680
R^2	0.573	0.207

Notes: Column 1 uses the monthly sum of daily union areas of NOAA HMS smoke plumes within the county exposure area (km^2-days), and Column 2 uses the monthly average county-level wildfire-specific smoke $PM_{2.5}$ concentration ($\mu g/m^3$). Both outcomes enter in levels, and zero outcomes are retained. The observational unit is the county-month. Both regressions use a balanced monthly panel for 2006–2020 of counties represented in the main birth-outcome analysis, without restricting individual months to those with reported births. The wildfire-smoke $PM_{2.5}$ data contain smoke days only; unreported county-days within this coverage period are assigned zero, and monthly means include all calendar days. The endogenous variable is log monthly natural-caused wildfire size, instrumented with monthly lightning counts. Fire areas and lightning counts are summed within each month, and weather variables are averaged before constructing logarithms and interactions. Each specification includes log other wildfire size, log precipitation, log humidity, log temperature, all pairwise and the three-way interactions among these logged weather variables, pressure, wind speed, county fixed effects, and year-month fixed effects. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. The fire-size measures add 10^{-6} km^2 before taking logarithms. The regressions are unweighted. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic, which is common to both regressions. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

D.2 Smoke Controls

Table D.3: Wildfires, Smoke Controls, and Birth Outcomes (IV, county-month level), 2006–2020

	Baseline	HMS Smoke	Smoke $PM_{2.5}$
	(1)	(2)	(3)
Panel A: Birth Weight (grams)			
Ln Natural-Caused Fire Size	-0.949***	-0.949***	-0.945***
	(0.333)	(0.334)	(0.334)
Ln HMS Smoke Area-Days		0.084*	
		(0.043)	
Ln Smoke $PM_{2.5}$			-0.153
			(0.528)
R^2	0.762	0.762	0.762
Panel B: Low Birth Weight per 1,000 Births			
Ln Natural-Caused Fire Size	0.296**	0.296**	0.300**
	(0.122)	(0.122)	(0.123)
Ln HMS Smoke Area-Days		-0.047***	
		(0.018)	
Ln Smoke $PM_{2.5}$			-0.129
			(0.193)
R^2	0.470	0.470	0.470
Panel C: LMP Gestational Age (weeks)			
Ln Natural-Caused Fire Size	-0.003	-0.003	-0.003
	(0.002)	(0.002)	(0.002)
Ln HMS Smoke Area-Days		0.001***	
		(0.0002)	
Ln Smoke $PM_{2.5}$			-0.003
			(0.003)
R^2	0.592	0.592	0.592
Panel D: Preterm Births per 1,000 Births			
Ln Natural-Caused Fire Size	0.305	0.305	0.290
	(0.200)	(0.200)	(0.201)
Ln HMS Smoke Area-Days		-0.014	
		(0.027)	
Ln Smoke $PM_{2.5}$			0.557**
			(0.283)
R^2	0.505	0.505	0.505
Obs.	98,136	98,136	98,136
F^{Eff}	46.69	47.33	45.24

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. All columns use the common 2006–2020 sample with both smoke measures and report IV estimates using lightning counts as the instrument. Column 1 reports the baseline estimate on this common sample. Column 2 adds the log of one plus HMS smoke area-days, constructed by summing the daily total area covered by HMS smoke within the county exposure area over the month. Its pre-log unit is km^2-days. Column 3 adds the log of one plus the monthly average wildfire smoke $PM_{2.5}$ concentration, measured in $\mu g/m^3$. All estimations include log other wildfire size, fire weather controls, pressure, wind speed, county fixed effects, and year-month fixed effects and are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Significance stars follow the conventional notation.

D.3 Twitter Sentiment

Table D.4: Wildfires and Twitter Sentiment (IV, county-day level), 2014–2020

	Std. Sentiment
	Baseline
	(1)
Ln Natural-Caused Fire Size	-0.088***
	(0.021)
County FE	X
Date FE	X
Fire Weather Controls	X
F^{Eff}	27.09
Mean (Dep. Var.)	0
SD (Dep. Var.)	1
Counties	576
Obs.	1,220,070
R^2	0.587

Notes: The dependent variable is the daily sentiment score constructed from Twitter posts (Baylis, 2020), standardized using the unweighted mean and standard deviation of the estimation sample. The observational unit is the county-day. The sample covers May 22, 2014, through December 31, 2020, and is restricted to counties with birth weight data and county-days with at least one geolocated post. Absent county-days remain missing. The table reports IV estimates of log natural-caused wildfire size using lightning counts as the instrument. The estimation includes log other wildfire size (combining identified human-caused wildfires and wildfires with undetermined or unclassified causes), fire weather controls, pressure, wind speed, county fixed effects, and date fixed effects and is weighted by county population. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics for the standardized outcome in the estimation sample. Significance stars follow the conventional notation.

D.4 Maternal Tobacco Use

Table D.5: Wildfires and Maternal Tobacco Use (IV, county-month level), 2007–2020

	Tobacco Use per 1,000 Births with Reported Tobacco Status
	Baseline
	(1)
Ln Natural-Caused Fire Size	1.289***
	(0.335)
County FE	X
Year-Month FE	X
Fire Weather Controls	X
F^{Eff}	30.26
Mean (Dep. Var.)	93
SD (Dep. Var.)	68
Counties	575
Obs.	72,149
R^2	0.898

Notes: The dependent variable is maternal tobacco use per 1,000 births with a reported tobacco status, defined as reported "Yes" births divided by reported "Yes" plus "No" births. Births coded "Unknown or Not Stated" or "Not Reported" are excluded from the denominator. Suppressed category counts are imputed as 5, the midpoint of the possible range from 1 to 9. The observational unit is the county-month. The 2007–2020 sample is restricted to county-months with birth data and reporting coverage of at least 95%. The table reports IV estimates of log natural-caused wildfire size using lightning counts as the instrument. The estimation includes log other wildfire size (combining identified human-caused wildfires and wildfires with undetermined or unclassified causes), fire weather controls, pressure, wind speed, county fixed effects, and year-month fixed effects and is weighted by the number of births with a reported tobacco status. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

D.5 Heterogeneous Effects

Table D.6: Heterogeneous Effects of Wildfires by County-Month Maternal Composition (IV, county-month level), 2003–2020

	Birth Weight (grams) (1)	LBW per 1,000 Births (2)	LMP Gestational Age (weeks) (3)	Preterm Births per 1,000 Births (4)	SW F (5)
Ln Natural-Caused Fire Size	-0.948**	0.158	-0.0021	0.225	229.7
	(0.403)	(0.137)	(0.0021)	(0.207)	
Ln Natural-Caused Fire Size × Black/African American	-0.817***	0.281***	-0.0026**	0.055	140.6
	(0.229)	(0.083)	(0.0010)	(0.098)	
Ln Natural-Caused Fire Size × Unmarried	0.387	0.171*	-0.0002	0.260	149.7
	(0.374)	(0.100)	(0.0020)	(0.163)	
Ln Natural-Caused Fire Size × Age of Mother	2.167***	-0.317***	0.0023	-0.108	128.1
	(0.469)	(0.109)	(0.0023)	(0.217)	
County FE	X	X	X	X	
Year-Month FE	X	X	X	X	
Fire Weather Controls	X	X	X	X	
Other Weather Controls	X	X	X	X	
Other Wildfire Size	X	X	X	X	
Mean (Dep. Var.)	3278	79	38.6	117	
SD (Dep. Var.)	76	26	0.3	36	
Counties	576	576	576	576	
Obs.	116,856	116,856	116,856	116,856	
R^2	0.761	0.473	0.587	0.502	

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. All columns report IV estimates. The demographic interaction variables are the Black/African American maternal share, unmarried maternal share, and average maternal age, standardized without weights within the common 2003–2020 outcome-estimation sample. The main wildfire-size coefficient is evaluated at zero for each standardized demographic modifier, corresponding to that sample mean. Each wildfire-size interaction is instrumented by the corresponding interaction between lightning counts and the demographic measure. All estimations include county fixed effects, year-month fixed effects, log other wildfire size, fire weather controls, pressure, wind speed, and demographic main effects. Estimations are weighted by the total number of births. Other wildfire size combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row, using the common outcome-estimation sample. The interaction coefficients describe heterogeneity across county-month maternal composition and do not identify individual-level subgroup effects. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

D.6 Direct Damage

Table D.7: Wildfires and Direct Damage (IV, county-month level), 2003–2020

	Buildings	Ln Prop. Loss p.c.	Ln Cont. Loss p.c.
	IV	IV	IV
	(1)	(2)	(3)
Ln Natural-Caused Fire Size	0.133**	0.020***	0.013***
	(0.057)	(0.007)	(0.005)
County FE	X	X	X
Year-Month FE	X	X	X
Weather Controls	X	X	X
F^{Eff}	57.81	57.81	57.81
Mean (Dep. Var.)	4.3	1.5	0.6
SD (Dep. Var.)	6.3	26.2	15.1
Counties	576	576	576
Obs.	116,856	116,856	116,856
R^2	0.198	0.350	0.204

Notes: "Buildings" is the number of buildings reported as directly involved in NFIRS fire incidents per 100,000 population. "Ln Prop. Loss p.c." and "Ln Cont. Loss p.c." are the log of one plus reported property and content losses in nominal dollars per resident, respectively. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, measured as buildings per 100,000 population and pre-log nominal dollars per resident for the two loss outcomes. The unconditional NFIRS construction retains all county-months, codes the absence of a matched NFIRS record as zero reported outcomes, and codes missing buildings-involved, property-loss, and content-loss fields within matched records as zero. NFIRS incidents are aggregated across all reported fires in a county-month and are not individually linked to the wildfires used to construct wildfire size. The sample is restricted to county-months with available average birth-weight data to match the health-analysis sample. The observational unit is the county-month. All columns report IV estimates of log natural-caused wildfire size using lightning counts as the instrument. All estimations control for log other wildfire size (combining identified human-caused wildfires and wildfires with undetermined or unclassified causes), fire weather, pressure, and wind speed and include county fixed effects and year-month fixed effects. Estimations are weighted by county population. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Significance stars follow the conventional notation.

D.7 Direct Injury

Table D.8: Wildfires and Casualties (IV, county-month level), 2003–2020

	All	Respiratory	Burn
	IV	IV	IV
	(1)	(2)	(3)
Ln Natural-Caused Fire Size	0.007*	0.001	0.002
	(0.004)	(0.002)	(0.001)
County FE	X	X	X
Year-Month FE	X	X	X
Weather Controls	X	X	X
F^{Eff}	57.81	57.81	57.81
Mean (Dep. Var.)	0.32	0.09	0.10
SD (Dep. Var.)	0.68	0.30	0.26
Counties	576	576	576
Obs.	116,856	116,856	116,856
R^2	0.178	0.126	0.145

Notes: All outcomes are NFIRS-reported civilian fire casualties per 100,000 population. "All" is total civilian fire casualties. "Respiratory" includes primary-symptom codes for smoke inhalation, hazardous-fume inhalation, and breathing difficulty or shortness of breath. "Burn" includes thermal, scald, chemical, and electrical burns, as well as burns accompanied by smoke inhalation. The unconditional NFIRS construction retains all county-months and codes the absence of a matched NFIRS record as zero reported casualties. NFIRS incidents are aggregated across all reported fires in a county-month and are not individually linked to the wildfires used to construct wildfire size. The sample is restricted to county-months with available average birthweight data. The observational unit is the county-month. All columns report IV estimates of log natural-caused wildfire size using lightning counts as the instrument. All estimations include log other wildfire size (combining identified human-caused wildfires and wildfires with undetermined or unclassified causes), fire weather controls, pressure, wind speed, county fixed effects, and year-month fixed effects, and are weighted by county population. Standard errors clustered by county are shown in parentheses. F^{Eff} is the effective first-stage F-statistic. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

E Fuel Treatment

E.1 Geographic, Temporal, and Size Distribution

Figures E.1 and E.2, along with Table E.1, summarize the temporal, geographic, and size distribution of the cleaned FACTS descriptive sample.

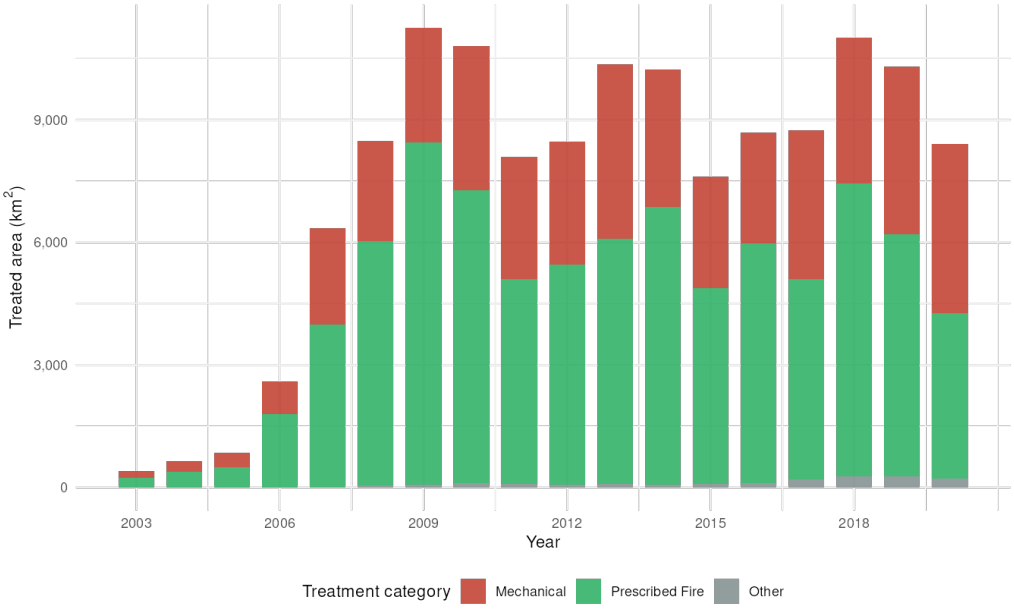


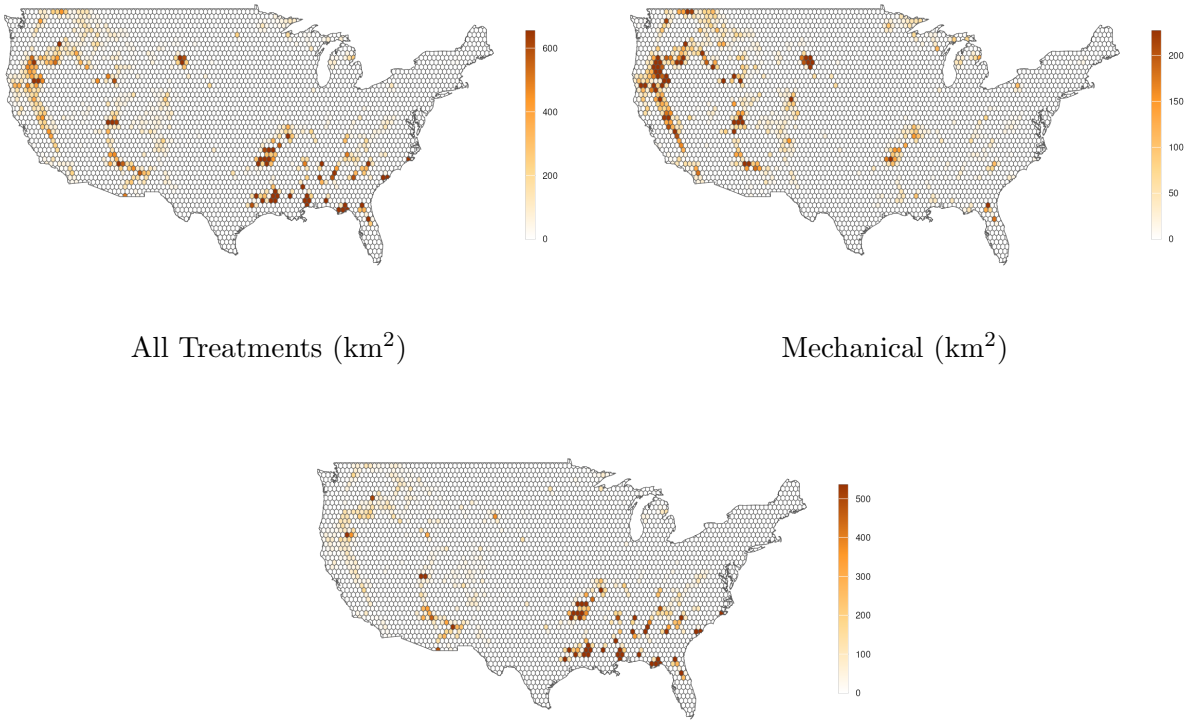
Figure E.1: Annual Fuel-Treatment Area by Type

Note: The figure uses completed, valid, positive-area FACTS treatment polygons in the contiguous US during 2003–2020. Treatment area is measured in km^2. Polygon areas are calculated in NAD83/Conus Albers (EPSG:5070).

Table E.1: Fuel-Treatment Area and Cost by Category and Treatment Type, 2003–2020

Treatment category	Treatment type	N	Total area (km^2)	Median area (km^2)	Mean \$/km^2	Total cost (million USD)
Fire	Broadcast Burn	33,358	75,611	0.587	32,514	2,120.1
	Jackpot Burn	1,097	660	0.153	30,753	13.8
	Machine Pile Burn	33,244	8,062	0.086	31,566	259.3
Mechanical	Biomass Removal	40,847	8,072	0.085	25,241	255.4
	Chipping	5,158	1,132	0.067	68,219	76.0
	Crushing	3,179	1,435	0.118	100,472	137.7
	Lop and Scatter	16,424	4,188	0.094	38,429	226.6
	Machine Pile	35,169	7,016	0.073	71,553	771.4
	Mastication/Mowing	294	65	0.053	87,098	12.6
	Thinning	114,418	25,347	0.093	41,672	1,542.2
Other	Grazing	1,330	1,187	0.237	18,268	36.5
	Other	3,573	488	0.061	42,442	31.4

Notes: This table summarizes completed, valid, positive-area FACTS fuel-treatment polygons in the contiguous US during 2003–2020, excluding administrative "Wildfire Non-Treatment" records and "Fire Use" records. The observational unit is a treatment polygon, and N counts the polygons used to calculate every reported area and cost statistic. Treatment categories are harmonized into fire, mechanical, and other treatments. The county measures of fuel-treatment area used in the policy regressions and benefit calculations include only mechanical and prescribed-fire treatments. Total area and median area are measured in km^2 and calculated in NAD83/Conus Albers (EPSG:5070). Mean cost per km^2 is the unweighted mean across treatment polygons within each category and treatment type. Mean cost and total cost are reported in nominal USD. Missing or outlier costs are imputed using observed polygon and administrative characteristics, while reported zero costs are retained. Cost statistics provide descriptive background and are not used in the birth-weight benefit calculation.



All Treatments (km^2)

Mechanical (km^2)

Prescribed Fire (km^2)

Figure E.2: Geographic Distribution of Fuel-Treatment Area

Note: The maps use the same sample as Figure E.1 and a regular hexagonal grid with 0.5 degrees between opposite edges. Detached islands are excluded, and the grid fully covers the mainland polygon. Hexagonal cells along the coast are clipped to the contiguous-US mainland coastline for display. Each map sums treatment area over 2003–2020. Each polygon's full area is assigned to the hexagon containing its projected centroid. Polygons with centroids outside the mainland are omitted from the maps.

Table E.2: Summary Statistics for Fuel-Treatment Area, 2003–2020

	Obs.	Mean	SD	Median	P90	Max	Zero (%)
Total	116,856	19.48	117.35	0	3.52	2,480.13	87.23
Mechanical	116,856	7.81	59.16	0	0.29	1,595.39	88.87
Prescribed Fire	116,856	11.68	81.87	0	0.59	1,717.14	88.28

Notes: This table reports unweighted summary statistics for beginning-of-month fuel-treatment area in the common birth-outcome estimation sample of Table 7. The observational unit is the county-month. The columns report observations (Obs.), mean, standard deviation (SD), median, ninetieth percentile (P90), maximum (Max), and the percentage of zeros. Areas are measured in km^2 before the logarithmic transformation used in the regressions. Total area is the sum of mechanical and prescribed-fire treatment areas completed before the first day of the month. Each treatment polygon contributes only the area inside the county boundary. Areas accumulate across completed treatment records without depreciation, and overlapping treatments can contribute more than once. Counties with no recorded treatment are retained with zero area. Zero (%) reports the percentage of county-month observations with zero area. The regressions add 10^{-6} km^2 before taking logarithms. These county-month measures differ from the polygon-level background statistics in Table E.1.

E.2 Fuel-Treatment Area and Estimation

Let d_t^0 denote the first day of month t, and let $\mathcal{C}_{ct}$ be the set of completed mechanical-treatment and prescribed-fire polygons that intersect county c and have completion date $\tau_j < d_t^0$. The beginning-of-month cumulative fuel-treatment area is

$$S_{ct} = \sum_{j \in \mathcal{C}_{ct}} a_{cj}, \tag{E.1}$$

where a_{cj} is the area of treatment polygon j inside the county boundary. Treatment areas accumulate across completed treatment records without depreciation, so overlapping treatment polygons can contribute more than once. When a treatment polygon crosses a county boundary, only the intersecting area inside county c enters S_{ct}, and no spatial buffer is used. Appendix Table E.4 applies a 10% annual depreciation rate.

The county-month model is

$$\begin{aligned} Y_{ct} = {} & \beta_1 \ln(NaturalFire_{ct}) + \beta_2 \ln(NaturalFire_{ct}) \times \ln(S_{ct}) + \gamma \ln(S_{ct}) \\ & + X_{ct}'\delta + \alpha_c + \lambda_t + \varepsilon_{ct}, \end{aligned} \tag{E.2}$$

where $NaturalFire_{ct}$ is natural-caused wildfire size aggregated over the county exposure area, X_{ct} contains weather controls and log other wildfire size (combining identified human-caused wildfires and wildfires with undetermined or unclassified causes), α_c is a county fixed effect, and λ_t is a year-month fixed effect. The implementation adds 10^{-6} to the treated area within the county before taking logs, but Equation E.2 omits this constant. Log natural-caused wildfire size and its interaction with the logged treatment area are instrumented by lightning counts and their interaction with the logged treatment area, respectively. The main effect of logged treatment area is included but is not instrumented. The interaction identifies conditional heterogeneity in the wildfire-size effect only if any direct effect of lightning on birth outcomes does not vary systematically with the preexisting treatment area. Because treatment placement is nonrandom, the interaction specification does not identify the causal effect of increasing treatment area.

E.3 Separated Treatment-Type Estimates

Table E.3: Fuel-Treatment Area and Wildfire Effects on Birth Outcomes by Treatment Type (IV, county-month level), 2003–2020

	Birth Weight (grams) (1)	LBW per 1,000 Births (2)	LMP Gestational Age (weeks) (3)	Preterm Births per 1,000 Births (4)	SW F (5)
Panel A: Mechanical Area					
Ln Natural-Caused Fire Size	-0.748***	0.229**	-0.002	0.424**	222.3
	(0.237)	(0.097)	(0.002)	(0.194)	
Ln Natural-Caused Fire Size × Ln Mechanical Area	0.053**	-0.009	0.0002	0.014	236.5
	(0.022)	(0.008)	(0.0001)	(0.016)	
R^2	0.764	0.473	0.585	0.501	
Panel B: Prescribed-Fire Area					
Ln Natural-Caused Fire Size	-0.851***	0.229**	-0.003*	0.462**	245.6
	(0.252)	(0.097)	(0.002)	(0.199)	
Ln Natural-Caused Fire Size × Ln Prescribed-Fire Area	0.048**	-0.010	0.0001	0.023	256.6
	(0.020)	(0.007)	(0.0002)	(0.018)	
R^2	0.763	0.473	0.585	0.500	
County FE	X	X	X	X	
Year-Month FE	X	X	X	X	
Weather Controls	X	X	X	X	
FT Area	X	X	X	X	
Mean (Dep. Var.)	3278	79	38.6	117	
SD (Dep. Var.)	76	26	0.3	36	
Counties	576	576	576	576	
Obs.	116,856	116,856	116,856	116,856	

Notes: This table reports separate regressions using beginning-of-month mechanical treatment area (Panel A) and prescribed-fire treatment area (Panel B) inside the county boundary in place of the combined measure in Table 7. Each regression includes only the indicated treatment-type measure and its interaction with wildfire size. Both area measures sum treatments completed before the first day of the month. Treatment polygons crossing county boundaries are allocated by their area inside each county, and no spatial buffer is used. The implementation adds 10^{-6} km^2 to the area for each treatment type before taking its logarithm. The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. The endogenous variables are log natural-caused wildfire size and its interaction with the logged area for the treatment type. The instruments are lightning counts and lightning counts interacted with the logged treatment area. Each specification includes county fixed effects, year-month fixed effects, the main effect of logged treatment area, log other wildfire size (combining identified human-caused wildfires with wildfires of undetermined or unclassified causes), fire weather controls, pressure, and wind speed and is weighted by the total number of births. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row, using the common outcome-estimation sample. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

E.4 Decay-Weighted Robustness

Table E.4: Decay-Weighted Fuel-Treatment Area and Wildfire Effects on Birth Outcomes (IV, county-month level), 2003–2020

	Birth Weight (grams) (1)	LBW per 1,000 Births (2)	LMP Gestational Age (weeks) (3)	Preterm Births per 1,000 Births (4)	SW F (5)
Ln Natural-Caused Fire Size	-0.830***	0.226**	-0.003*	0.416**	235.0
	(0.255)	(0.097)	(0.002)	(0.178)	
Ln Natural-Caused Fire Size × Ln Decay-Weighted FT Area	0.052**	-0.011	0.0001	0.018	234.1
	(0.021)	(0.008)	(0.0001)	(0.015)	
County FE	X	X	X	X	
Year-Month FE	X	X	X	X	
Weather Controls	X	X	X	X	
FT Area	X	X	X	X	
Mean (Dep. Var.)	3278	79	38.6	117	
SD (Dep. Var.)	76	26	0.3	36	
Counties	576	576	576	576	
Obs.	116,856	116,856	116,856	116,856	
R^2	0.763	0.473	0.585	0.500	

Notes: This table is identical to Table 7 except that the area measure weights completed treatments using a 10% annual depreciation rate rather than retaining their full area indefinitely. The area is measured at the beginning of the month, treatment polygons crossing county boundaries are allocated by their area inside each county, and no spatial buffer is used. The implementation adds 10^{-6} km^2 before taking the logarithm of treatment area. Each specification controls for log other wildfire size, which combines identified human-caused wildfires with wildfires of undetermined or unclassified causes. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row, using the common outcome-estimation sample. Standard errors clustered by county are shown in parentheses. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, using the outcome definitions in Table 7. Significance stars follow the conventional notation.

E.5 Other Outcomes by Fuel-Treatment Area

Table E.5: Fuel-Treatment Area and Wildfire Effects on Smoke Exposure (IV, county-day level), 2006–2020

	Wildfire Smoke $PM_{2.5}$ (1)	HMS Smoke Area (2)	SW F (3)
Ln Natural-Caused Fire Size	0.242***	441.973***	84.0
	(0.038)	(67.892)	
Ln Natural-Caused Fire Size × Ln FT Area	-0.029***	-46.336***	82.6
	(0.005)	(8.274)	
Mean (Dep. Var.)	0.395	591.802	
SD (Dep. Var.)	2.731	2108.263	
R^2	0.099	0.099	
County FE	X	X	
Date FE	X	X	
Weather Controls	X	X	
FT Area	X	X	
Counties	576	576	
Obs.	3,155,904	3,155,904	

Notes: Column 1 uses daily county-level wildfire-specific smoke $PM_{2.5}$ concentration ($\mu g/m^3$), and Column 2 uses the daily union area of NOAA HMS smoke plumes within the county exposure area (km^2). Both outcomes enter the regressions in levels, and zero outcomes are retained. The official wildfire-smoke $PM_{2.5}$ data cover January 1, 2006, through December 31, 2020, and contain smoke days only. Unreported county-days within this coverage period are assigned zero. The observational unit is the county-day, and both regressions use the same sample. Ln FT Area denotes the log of the sum of mechanical and prescribed-fire treatment areas completed before the observation day within the county. Treatment polygons crossing county boundaries are allocated by their area inside each county, no spatial buffer is used, and 10^{-6} km^2 is added before taking the logarithm. The endogenous variables are log natural-caused wildfire size and its interaction with the logged treatment area. The instruments are lightning counts and lightning counts interacted with the logged treatment area. Each specification includes the main effect of logged treatment area, log other wildfire size (combining identified human-caused wildfires with wildfires of undetermined or unclassified causes), log of one plus precipitation, log humidity, log temperature, all pairwise and the three-way interactions among these logged weather variables, pressure, wind speed, county fixed effects, and date fixed effects. The regressions are unweighted. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, measured in the outcome units specified above. The wildfire-size main coefficient is evaluated where logged treatment area equals zero (approximately 1 km^2 of treatment area). Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row. Both regressions share the first-stage sample. Significance stars follow the conventional notation.

Table E.6: Fuel-Treatment Area and Wildfire Effects on Twitter Sentiment (IV, county-day level), 2014–2020

	Std. Twitter Sentiment (1)	SW F (2)
Ln Natural-Caused Fire Size	-0.054***	65.1
	(0.013)	
Ln Natural-Caused Fire Size × Ln FT Area	0.005**	64.3
	(0.002)	
County FE	X	
Date FE	X	
Weather Controls	X	
FT Area	X	
Mean (Dep. Var.)	0	
SD (Dep. Var.)	1	
Counties	576	
Obs.	1,220,070	
R^2	0.580	

Notes: The dependent variable is the county-day mean VADER sentiment score, standardized using the unweighted mean and standard deviation in the common estimation sample. Higher values indicate more positive sentiment. The Twitter sentiment data begin on May 22, 2014, and the estimation sample ends on December 31, 2020. County-days absent from the source data remain missing rather than being assigned a neutral score, and the sample is restricted to counties observed in the main birth-outcome data. The observational unit is the county-day. Ln FT Area denotes the log of the sum of mechanical and prescribed-fire treatment areas completed before the observation day within the county. Treatment polygons crossing county boundaries are allocated by their area inside each county, no spatial buffer is used, and 10^{-6} km^2 is added before taking the logarithm. The endogenous variables are log natural-caused wildfire size and its interaction with the logged treatment area. The instruments are lightning counts and lightning counts interacted with the logged treatment area. The estimation includes the main effect of logged treatment area, log other wildfire size (combining identified human-caused wildfires with wildfires of undetermined or unclassified causes), log of one plus precipitation, log humidity, log air temperature, their pairwise and three-way interactions, pressure, wind speed, county fixed effects, and date fixed effects, and is weighted by county population. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics for the standardized outcome in the common estimation sample. Significance stars follow the conventional notation.

Table E.7: Fuel-Treatment Area and Wildfire Effects on Maternal Tobacco Use (IV, county-month level), 2007–2020

	Tobacco Use per 1,000 Births with Reported Tobacco Status (1)	SW F (2)
Ln Natural-Caused Fire Size	0.980***	164.1
	(0.286)	
Ln Natural-Caused Fire Size × Ln FT Area	-0.065**	129.4
	(0.029)	
County FE	X	
Year-Month FE	X	
Weather Controls	X	
FT Area	X	
Mean (Dep. Var.)	93	
SD (Dep. Var.)	68	
Counties	575	
Obs.	72,149	
R^2	0.895	

Notes: The dependent variable is maternal tobacco use during pregnancy per 1,000 births with a reported tobacco status, defined as reported "Yes" births divided by reported "Yes" plus "No" births. Births coded "Unknown or Not Stated" or "Not Reported" are excluded from the denominator, and suppressed category counts are imputed as 5, the midpoint of the possible range from 1 to 9. The observational unit is the county-month. The 2007–2020 sample is restricted to county-months with reporting coverage of at least 95%. Ln FT Area denotes the log of the sum of mechanical and prescribed-fire treatment areas completed before the first day of the month within the county. Treatment polygons crossing county boundaries are allocated by their area inside each county, no spatial buffer is used, and 10^{-6} km^2 is added before taking the logarithm. The endogenous variables are log natural-caused wildfire size and its interaction with the logged treatment area. The instruments are lightning counts and lightning counts interacted with the logged treatment area. The estimation includes the main effect of logged treatment area, log other wildfire size (combining identified human-caused wildfires with wildfires of undetermined or unclassified causes), fire weather controls, pressure, wind speed, county fixed effects, and year-month fixed effects, and is weighted by the number of births with a reported tobacco status. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.

E.6 Back-of-the-Envelope Birth-Weight Benefit Calculation

This calculation values the birth-weight improvement associated with an expansion of completed mechanical and prescribed-fire treatment flows through the severity channel. The birth-weight specification is the first column of Table 7. Let $\mathcal{C}^{FT}$ denote the 108 counties represented in both the birth-outcome estimation sample and the county treatment-area panel. Five of these counties have no recorded completed mechanical or prescribed-fire treatments throughout 2003–2020 and contribute zero benefits. Let $\mathcal{M}^{FT}$ denote the 21,612 county-month observations for counties in $\mathcal{C}^{FT}$ that belong to the birth-outcome estimation sample. This set covers January 2003–December 2020 for 95 counties and January 2014–December 2020 for the remaining 13 counties. Let $T = 18$ denote the number of years from 2003 to 2020.

The initial treatment area at the beginning of January 2003 S_{c0} is held fixed. Expanding subsequent treatment flows by a proportion q produces the counterfactual beginning-of-month treatment area

$$S_{ct}^{q} = S_{c0} + (1 + q)(S_{ct} - S_{c0}). \tag{E.3}$$

Each additional treatment is assigned to the same county and completion month as its observed counterpart. The calculation assigns no additional treatment to the 468 sample counties outside the county treatment-area panel. Additional treatments remain in the area measure in subsequent months without depreciation and can therefore affect later birth cohorts within the sample period. Figure E.3 varies q from zero to one, corresponding to increases of 0–100% in annual treatment area. This is an increase in treatment flows, not a uniform percentage increase in the preexisting treatment area.

The horizontal axis varies the average annual natural-caused wildfire area summed over $\mathcal{M}^{FT}$ from 0.1 to 4,000 km^2 on a logarithmic scale. For a scenario area A, each fire area in $\mathcal{M}^{FT}$ is multiplied by $A/\bar{A}$, where $\bar{A} = T^{-1} \sum_{(c,t)\in\mathcal{M}^{FT}} F_{ct}$. Thus, $F_{ct}(A) = F_{ct}A/\bar{A}$. This construction preserves zero-fire observations and the relative spatial and seasonal pattern of exposure. Treatment expansion leaves fire area unchanged within each scenario.

Let $g(F) = \ln(F + \varepsilon)$ and $h(S) = \ln(S + \varepsilon)$, where $\varepsilon = 10^{-6}$ and both areas are measured in km^2. Using the birth-weight interaction coefficient $\widehat{\beta}_2$ from Equation E.2, the severity-channel gain in grams per birth is

$$\Delta BW_{ct}(A, q) = \widehat{\beta}_2 \left[h(S_{ct}^{q}) - h(S_{ct})\right] \left[g(F_{ct}(A)) - g(0)\right]. \tag{E.4}$$

This expression measures how additional treatment changes the fitted birth-weight differ-

ence between wildfire exposure and no wildfire. The main effect of logged treatment area cancels in this comparison, and the gain is zero in county-months without natural-caused wildfire. The calculation uses birth-month exposure without adding lagged effects. Applying the interaction to a treatment expansion assumes that the estimated relationship across preexisting treatment areas remains valid under that expansion. This assumption is not identified by the interaction design.

Medical-cost savings follow the per-gram valuation used by East et al. (2023).[31] Their \$8.29 benchmark in 2011 dollars becomes \$9.54 in 2020 dollars, denoted by v_M. Let N_{ct}^{BW} denote births with reported birth weight. Average annual medical-cost savings are

$$B_M(A,q) = \frac{1}{T} \sum_{(c,t)\in\mathcal{M}^{FT}} N_{ct}^{BW}\, \Delta BW_{ct}(A,q)\, v_M. \tag{E.5}$$

The additional earnings calculation uses the birth-weight elasticity of permanent labor income, $\theta = 0.127$, reported in Table 4, Panel B, column 2 of Bharadwaj et al. (2018). A first-order approximation converts the proportional birth-weight gain into a proportional earnings gain. This assumes that the elasticity estimated from Swedish twins' income at ages 35–45 applies to lifetime earnings in the target population. The baseline present value of lifetime money earnings, L, is \$663,222 in 2020 dollars. This value comes from the all-sexes market-productivity estimate at birth in Table 2 of Grosse et al. (2019), using a 3% discount rate and 0.5% annual real earnings growth. Their estimate incorporates survival probabilities. Dividing their 2016-dollar value of \$758,954 by the employer-compensation multiplier of 1.234 removes employer-paid benefits before conversion using the Consumer Price Index for All Urban Consumers (CPI-U). With BW_{ct} denoting observed county-month average birth weight, the earnings benefit is

$$B_E(A,q) = \frac{1}{T} \sum_{(c,t)\in\mathcal{M}^{FT}} N_{ct}^{BW}\, \theta \frac{\Delta BW_{ct}(A,q)}{BW_{ct}}\, L. \tag{E.6}$$

Benefits per annual birth cohort are calculated by summing gains over births with reported birth weight in $\mathcal{M}^{FT}$ and dividing by the 18 years from 2003 to 2020. No benefits are imputed for county-months outside this set. The earnings component is the present value of that cohort's future earnings gain at birth, not income received within the birth year. The calculation excludes benefits to cohorts born after 2020 and does not discount all cohorts to a common program start date.

These valuations extrapolate medical-cost estimates for infants weighing 2,000–2,100 grams and earnings estimates for Swedish twins aged 35–45. Applying them across the

[31] The per-gram calculation is described in footnote 52 of that study.

current birth-weight distribution and working life may overstate benefits if marginal gains are smaller at higher birth weights or other ages. The results also depend on the additive constant defining the no-fire comparison, no depreciation of past treatment areas, and extrapolation beyond observed wildfire areas and treatment expansions. The figure reports point estimates conditional on these assumptions, excluding benefits from reduced burned area, property damage, and other health outcomes.

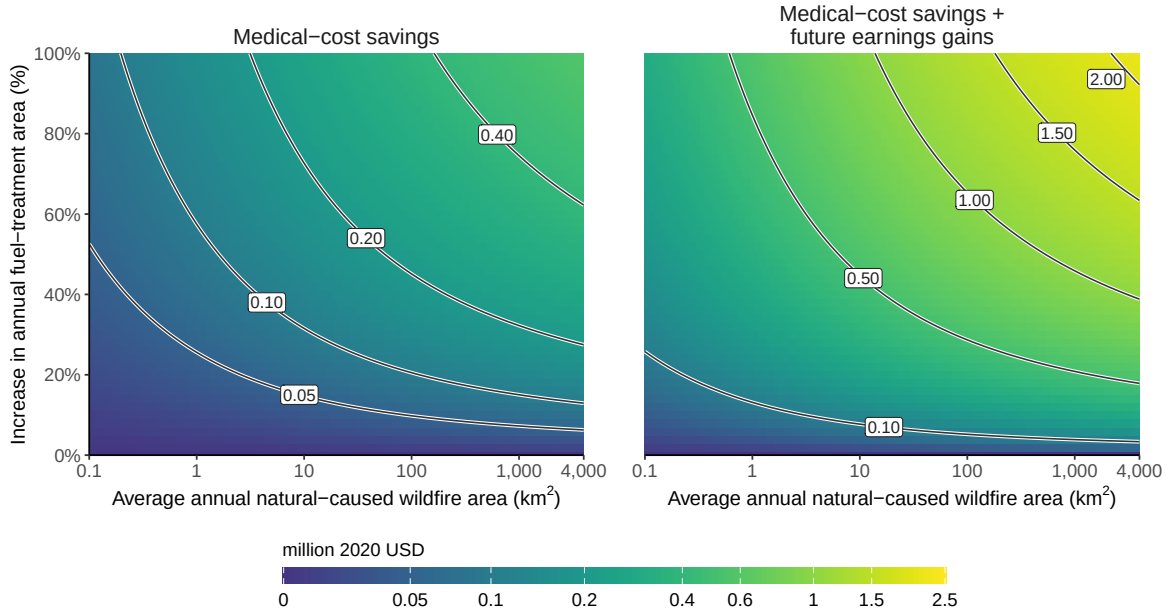


Figure E.3: Annual Birth-Weight Benefits of Fuel-Treatment Expansion

Note: The figure reports illustrative severity-channel benefits for 108 counties represented in both the county treatment-area panel and the birth-outcome estimation sample. Calculations use 21,612 county-months covering 2003–2020 for 95 counties and 2014–2020 for 13 counties. The horizontal axis shows natural-caused wildfire area in km^2 summed over these observations and divided by 18, on a logarithmic scale. The vertical axis shows the percentage increase in annual treatment area. These additions remain in the treatment-area measure in subsequent months, while the initial area remains fixed. The left panel reports medical-cost savings. The right panel adds lifetime labor earnings discounted to birth. Benefits are summed over births with reported birth weight in these observations and divided by 18. No benefits are imputed for county-months outside the estimation sample. Colors and labeled contours show millions of 2020 dollars per annual birth cohort. Both panels use the same nonlinear color scale, which expands differences at low benefit values. Contours connect scenarios with equal benefits. The shading represents benefit levels, not confidence intervals. The figure uses point estimates and does not display uncertainty in the estimated interaction or the external valuations.

E.7 Exposure-Timing Heterogeneity

Table E.8: Fuel-Treatment Area and Wildfire Effects on Birth Outcomes by Exposure Timing (IV, county-month level), July 2003–December 2020

	Birth Weight		LMP Gestational Age		
	Birth Weight (grams) (1)	LBW per 1,000 Births (2)	LMP Gestational Age (weeks) (3)	Preterm Births per 1,000 Births (4)	SW F (5)
Ln Natural-Caused Fire Size: Birth Month	-0.660	0.220	-0.003	0.620	266.8
	(0.538)	(0.168)	(0.004)	(0.469)	
Ln Natural-Caused Fire Size × Ln FT Area: Birth Month	0.094**	-0.007	0.0002	0.069*	205.0
	(0.044)	(0.014)	(0.0003)	(0.036)	
Ln Natural-Caused Fire Size: 3 Months Before Birth	-0.853*	-0.066	-0.005	0.053	152.5
	(0.518)	(0.157)	(0.003)	(0.414)	
Ln Natural-Caused Fire Size × Ln FT Area: 3 Months Before Birth	0.083**	-0.022*	0.0003	0.028	193.0
	(0.036)	(0.012)	(0.0002)	(0.025)	
Ln Natural-Caused Fire Size: 6 Months Before Birth	-0.416	-0.186	-0.005	0.369	132.0
	(0.621)	(0.192)	(0.005)	(0.648)	
Ln Natural-Caused Fire Size × Ln FT Area: 6 Months Before Birth	0.139**	0.001	0.0004	0.082*	123.5
	(0.055)	(0.017)	(0.0004)	(0.045)	
Sum of FT Interactions	0.317**	-0.027	0.001	0.179*	
	(0.128)	(0.038)	(0.001)	(0.103)	
Joint Wald p-value: FT Interactions	0.059	0.091	0.101	0.171	
Equality Wald p-value: FT Interactions	0.162	0.192	0.131	0.084	
County FE	X	X	X	X	
Year-Month FE	X	X	X	X	
Current and Lagged Fire/Weather Controls	X	X	X	X	
Exposure-Month FT Area Main Effects	X	X	X	X	
Mean (Dep. Var.)	3277	79	38.6	117	
SD (Dep. Var.)	76	26	0.3	36	
Counties	576	576	576	576	
Obs.	113,736	113,736	113,736	113,736	
R^2	0.746	0.472	0.563	0.493	

Notes: The dependent variables are county average birth weight (grams), low birth weight infants per 1,000 births with reported birth weight, county average LMP gestational age (weeks), and births below 37 completed LMP weeks per 1,000 births with reported LMP gestational age. The observational unit is the county-month. Log natural-caused wildfire size in the birth month and three and six months before birth is interacted with logged mechanical plus prescribed-fire treatment area inside the county boundary at the corresponding exposure month. Each treatment-area measure sums treatments completed before the first day of the corresponding exposure month. The implementation adds 10^{-6} km^2 before taking the logarithm of each exposure-month treatment area. The three exposure-month main effects of logged treatment area are included but omitted from the table. The six endogenous variables are the three fire terms and their three interactions with treatment area. The instruments are the corresponding lightning counts and interactions of lightning counts with logged treatment area. All estimations include county fixed effects, year-month fixed effects, contemporaneous and corresponding lagged log other wildfire size (combining identified human-caused wildfires with wildfires of undetermined or unclassified causes), fire weather controls, pressure, and wind speed. Estimations are weighted by the total number of births. The Sum of FT Interactions row adds the three interaction coefficients. The joint Wald test evaluates whether all three interactions equal zero, and the equality Wald test evaluates whether the three interactions are equal. Standard errors clustered by county are shown in parentheses. The final column reports each Sanderson–Windmeijer (SW) F-statistic on the corresponding endogenous-variable coefficient row, using the common outcome-estimation sample. The sample starts in July 2003 so that the six-month lag and the corresponding treatment area are observed after the county treatment-area panel begins. Mean (Dep. Var.) and SD (Dep. Var.) are unweighted statistics in the common estimation sample, in the outcome units stated above. Significance stars follow the conventional notation.